

Buried Box Design Example to BS EN 1997-1 and PD 6694-1:2011

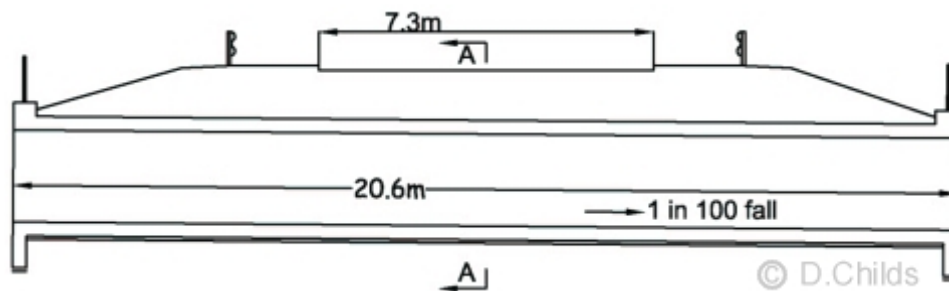
The following example is not a definitive solution for the design of a box culvert, but provides one solution to demonstrate the principles involved when considering design standards PD 6694-1:2011 and BS EN 1997-1. The example is presented to promote comment and discussion.

Index

- 1) [Brief](#)
 - 2) [Loading](#)
 - 3) [Analysis](#)
 - 4) Load Case Tables [B.1](#) [B.2](#) [B.3](#) [B.4](#) [B.5](#) [B.6](#)
 - 5) [Reinforcement Design](#)
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Brief

Design the in-situ concrete culvert shown below to carry LM1 and LM3 vehicle SV196. Analyse the culvert using a unit strip method. The bridge site is located south east of Oxford (to establish the range of shade air temperatures).

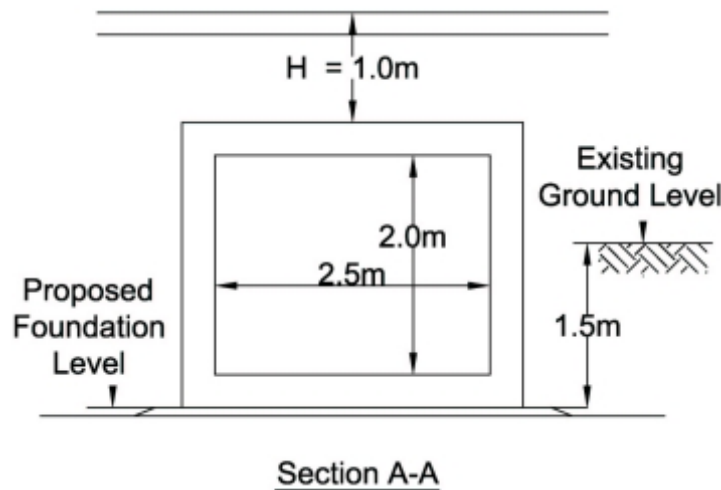


The ground investigation report shows the founding strata to be a well-drained cohesionless soil having an angle of shearing resistance (ϕ') = 32° and classified as 'not hard' material. Density of the foundation material is 19kN/m^3 .

Maximum height of ground water is 1m above the invert level of the culvert.

Maximum height of water inside the culvert is 1.8m above the invert level of the culvert.

Proposed foundation level is 1.5m below existing ground level.



Backfill material will be Class 6N with a density (γ) = 19kN/m^3 .

Take the angle of shearing resistance for Class 6N backfill as $\phi' = 35^\circ$

Fill over the culvert consists of 0.5m of Class 6N overlaid with 0.3m of road sub-base with a density (γ) = 20kN/m^3 and 0.2m of carriageway construction with a density (γ) = 24kN/m^3

The initial design will be based on using C32/40 concrete to BS 8500 and Grade B500B reinforcement to BS 4449.

Loading

Consider loading on 1m strip of the culvert.

i. Self Weight

Assume a wall thickness of 300mm and concrete density (γ) = 25kN/m^3

Nominal total load = $25 \times [(3.1 \times 2.6) - (2.5 \times 2.0)] = 76.5\text{kN}$

ii. Fill Over Box

Nominal load per metre = $19 \times 0.5 + 20 \times 0.3 = 15.5\text{kN/m}$

iii. Road Construction

Consider the top 200mm as road construction.

Nominal load per metre = $24 \times 0.2 = 4.8\text{kN/m}$

iv. Settlement

The MCHW Specification Series 600 ensures that any soft spots are removed from the foundation and replaced with suitable material. This will ensure that any differential settlement for a relatively short span box will be negligible. Also a suitable factor of safety will be applied to the allowable bearing pressure when checking the maximum

bearing pressure under the box to ensure that settlement will be minimal.

v. Load Model 1: UDL + Tandem Systems

Number of notional lanes = $n_1 = \text{Int}(w/3) = \text{Int}(7.3/3) = 2$

Notional Lane Width = 3.0m

Width of remaining area = $7.3 - 2 \times 3.0 = 1.3\text{m}$

UDL in Lane 1 = $a_{q1}q_{1k} = 0.61 \times 9 = 5.5\text{kN/m}^2$

UDL in Lane 2 = $a_{q2}q_{2k} = 2.2 \times 2.5 = 5.5\text{kN/m}^2$

UDL in remaining area = $a_{qr}q_{rk} = 2.2 \times 2.5 = 5.5\text{kN/m}^2$

TS in Lane 1 = $Q_{1k} = 300\text{kN}$

TS in Lane 2 = $Q_{2k} = 200\text{kN}$

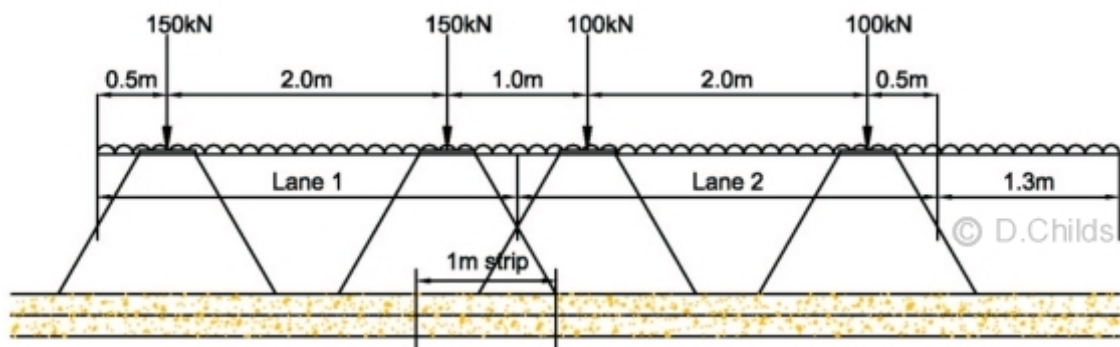
Contact patch area = $400 \times 400\text{mm}$

Dispersed area on top of box = $400 + 2 \times 1000 \times \tan 30^\circ = 1555 \times 1555\text{mm}$

Edge of carriageway to headwall at end of structure $\approx 6.1\text{m} > (1000 \times \tan 30^\circ + 300 / 2) = 727\text{mm}$.

Also there are no longitudinal joints in the structure, hence full dispersal can be considered.

Transverse Dispersal:



From PD 6694-1:2011 Figure 11.

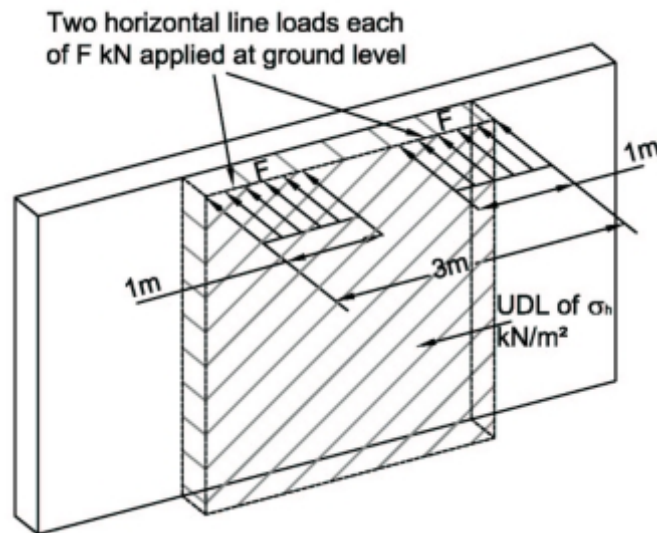
Load per metre where dispersion zones overlap = $bW_1/L_1 + aW_2/L_2 = 1 \times 150 / 1.555 + (1.555 - 1.0) \times 100 / 1.555 = 132.2\text{kN/m}$

Longitudinal Dispersal:

Dispersal zone width for each axle = 1.555m

Patch load for each axle = $132.2 / 1.555 = 85.0\text{kN/m}$

vi. Horizontal Surcharge Model for LM1, LM2 & LM3



Horizontal line load $F = 330K_d D_f$ where $K_d = K_a$ or K_0 and $D_f = 1$.

Reduction factor $= (1 - H_c / 2)^2 = (1 - 1 / 2)^2 = 0.25$

Two line loads are applied so for a 3m lane width the load on a 1m wide strip $= 2 \times 0.25 \times 330K_d / 3 = 55K_d \text{ kN}$.

K_a or K_0 are obtained from the appropriate load case Table in Annex B.

UDL for LM1 & LM2 $= 20K_d R = 20K_d \text{ (kN/m}^2\text{)}$

UDL for LM3 $= 30K_d R = 30K_d \text{ (kN/m}^2\text{)}$

vii. **Load Model 2**

Wheel load $= 200 \text{ kN}$

Contact patch area $= 400 \times 400 \text{ mm}$

Wheel spacing on axle $= 2.0 \text{ m} \therefore$ dispersal zones do not overlap (see Figure 11 diagram above).

Longitudinal and Transverse dispersal of wheel load $= 400 + 2 \times 1000 \times \tan 30^\circ = 1555 \times 1555 \text{ mm}$

Patch load for each wheel $= 200 / 1.555^2 = 83.25 \text{ kN/m}$

viii. **Load Model 3: SV196 Vehicle**

Axle 1 load $= 100 \text{ kN}$ Dynamic Amplification Factor $= 1.2$ Wheel Load $= 1.2 \times 50 = 60 \text{ kN}$

Axles 2 & 3 loads $= 180 \text{ kN}$ Dynamic Amplification Factor $= 1.1$ Wheel Load $= 1.1 \times 90 = 99 \text{ kN}$

Trailer Axle loads $= 165 \text{ kN}$ Dynamic Amplification Factor $= 1.12$ Wheel Load $= 1.12 \times 82.5 = 92.4 \text{ kN}$

Contact patch area $= 350 \times 350 \text{ mm}$

Longitudinal and Transverse dispersal of wheel load $= 350 + 2 \times 1000 \times \tan 30^\circ = 1505 \times 1505 \text{ mm}$

Wheel spacing on axles $= 2.65 \text{ m}$

Loading in Lane 2 is taken as the Frequent loading from LM1 (gr.5 combination)

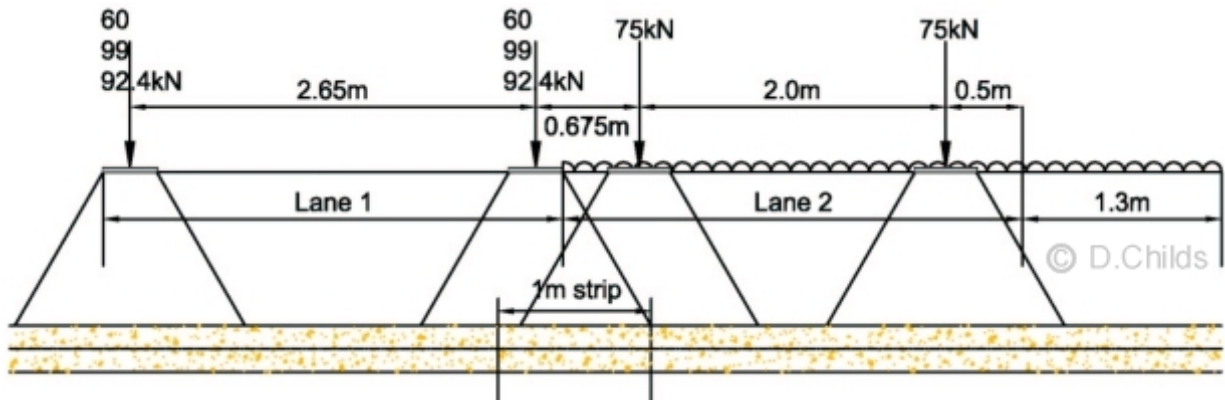
Frequent Value of Tandem System = $\psi_1 \times Q_{2k} \times \alpha Q_2$

TS in Lane 2 = $Q_{2k} = 200\text{kN}$

Frequent value of wheel load = $0.75 \times 100 \times 1.0 = 75\text{kN}$

Longitudinal and Transverse dispersal of TS2 wheel load = $400 + 2 \times 1000 \times \tan 30^\circ$
= $1555 \times 1555\text{mm}$

Transverse Dispersal:



Dispersal zones of adjacent SV and TS wheels overlap (see diagram above).

Consider 1m strip as shown then:

Patch load from SV196 Axle 1 = $60 / 1.505^2 = 26.5\text{kN/m}^2$

Patch load from SV196 Axle 2 = $99 / 1.505^2 = 43.7\text{kN/m}^2$

Patch load from SV196 Trailer Axle = $92.4 / 1.505^2 = 40.8\text{kN/m}^2$

Patch load from TS2 Axle = $(1.555 - 0.7) \times 75 / 1.555^2 = 26.5\text{kN/m}^2$

These patch loads are traversed across the structure at the appropriate axle spacings to obtain the worst effects.

ix. Critical Traffic Load

A [spreadsheet is available from this website](#) to analyse the moving load effects from the LM1, LM2 and LM3 traffic.

The [results from the spreadsheet](#) show that the critical load effects (bending moments and shear) are obtained under LM1 traffic.

x. Temperature Load

From NA to BS EN 1991-1-5:2003 Clause NA.2.2.1(b):

Span to width ratio = $X_{\text{clear}} / L_T = 2.5 / 20.6 = 0.12 (<0.2)$

Cover (fill + surfacing) = $1.0 > 0.6\text{m}$

Therefore the structure may be considered to be protected from climatic and operational temperature changes and temperature effects need not be considered for sections under the carriageway.

A [spreadsheet has been developed](#) to analyse temperature effects in a box structure

should it be required.

xi. **Braking and Acceleration Forces** to PD 6694-1:2011 Clause 10.2.8.2

$H_c < L_L \therefore$ the effects of braking and acceleration need to be considered.

For LM1: $Q_{lk} = 0.6\alpha_{Q1}(2Q_{1k}) + 0.1\alpha_{q1}q_{1k}w_1L$

$$180\alpha_{Q1} \leq Q_{lk} \leq 900\text{kN}$$

$$\alpha_{Q1} = \alpha_{q1} = 1$$

$$Q_{1k} = 300 \text{ kN}$$

$$q_{1k} = 9 \text{ kN/m}$$

$$\text{Lane width } w_1 = 3\text{m}$$

$$L = 3.1\text{m}$$

$$\text{Earth cover } 0.6\text{m} < H_c < L_L$$

$$\text{Reduction Factor } \eta = (L_L - H_c) / (L_L - 0.6) = (3.1 - 1.0) / (3.1 - 0.6) = 0.84$$

$$\text{Hence } Q_{lk} = 0.84 \times (360 + 8.37) = 309.4\text{kN}$$

PD 6694 says Q_{lk} should be applied directly to the top of the roof of the structure over a notional lane width, but may be assumed to be distributed structurally over a length of L_j provided account is taken of the tendency for plan rotation.

Taking an extreme condition with braking/acceleration in each lane at maximum eccentricity from the centre of the carriageway then:

$$\text{Rotation moment} = 2 \times 309.4 \times 2.15 = 1330\text{kNm}$$

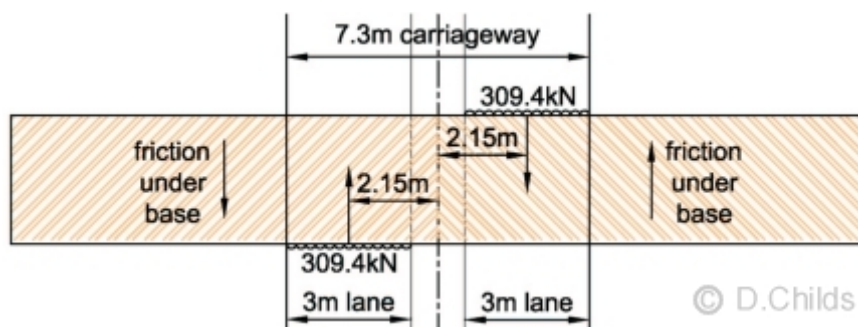
Assume rotation resisted by friction under the base then:

$$\text{Approximate volume of road construction} = 7.3 \times 3.1 \times 0.2 = 5\text{m}^3$$

$$\text{Approximate volume of sub-base} = 7.3 \times 3.1 \times 0.3 = 7\text{m}^3$$

$$\text{Approximate volume of 6N fill over box assuming 1:3 embankment slopes} = (20.6 - 3) \times 1.0 \times 3.1 - (5 + 7) = 43\text{m}^3$$

$$\text{Nominal weight of box and fill over} = 76.5 \times 20.6 + 3 \times 24 + 7 \times 20 + 43 \times 19 = 2605\text{kN}$$



Under SLS limit state γ_F and $\gamma_M = 1.0$ so:

$$\text{Friction force available under each half of box} = 2605 / 2 \times \tan 32^\circ = 814\text{kN}$$

$$\text{Friction moment available for resisting rotation} = 2 \times 814 \times 20.6 / 4 = 8384\text{kNm} \gg 1330\text{kNm} \therefore \text{OK.}$$

No allowance has been made for the earth pressures on the side of the box in this over-

simplified calculation; a net resistance of at-rest pressure less active pressure ($K_0 - K_a$) could also be considered. However the calculation justifies adopting the allowance for the in-plane rigidity of the roof and walls of in-situ box in PD 6694-1:2011 Clause 10.2.8.2 by distributing the braking/acceleration force over L_j , and as there are no joints, $L_j = 20.6\text{m}$
 Braking/acceleration force = $309.4 / 20.6 = 15\text{kN/m}$

For LM3: $Q_{lk,s} = \delta\omega$

$\delta = 0.25$ for SV196 vehicle

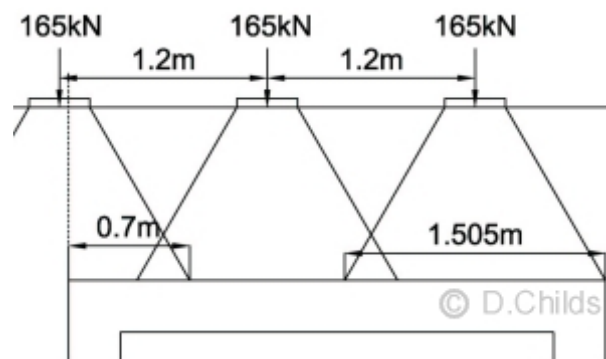
ω = Basic Axle Loads = 100kN, 180kN & 165kN

$Q_{lk,s}$ (axle 1) = $0.25 \times 100\text{kN} = 25\text{kN}$

$Q_{lk,s}$ (axles 2 & 3) = $0.25 \times 180\text{kN} = 45\text{kN}$

$Q_{lk,s}$ (axles 4 to 12) = $0.25 \times 165\text{kN} = 41.25\text{kN}$

The maximum vertical load that can be positioned over the box is from the trailer axles. Position the edge of the dispersed axle load zone from the leading trailer axle at the edge of the culvert.



PD 6694-1:2011 Clause 10.2.8.2 says: "The total braking or acceleration force applied to the top of the roof of a buried structure need not be taken as greater than the friction force that can be generated between the earth and the roof."

Maximum vertical load from LM3 on a unit width of box = $40.8 \times (2 \times 1.505 + 0.7) = 151.4\text{kN}$.

Vertical load from fill above culvert = $(15.5 + 4.8) \times 3.1 = 62.9\text{kN}$

Assuming a coefficient of friction of $\tan 30^\circ$ then:

Maximum friction that can be generated on a metre width between the earth and the roof is $(151.4 + 62.9)\tan 30^\circ = 124\text{kN}$

Total braking force generated by SV196 vehicle = $25 + 2 \times 45 + 9 \times 41.25 = 486\text{kN}$

Using the in-plane rigidity of the roof and walls as proven above then

braking/acceleration force = $486 / 20.6 = 24\text{kN/m} < 124\text{kN} \therefore$ use 24kN/m

xii. Centrifugal & Other Transverse Forces

Assuming a straight carriageway over the structure then there will be no centrifugal forces acting.

Other transverse forces are taken as 50% of the longitudinal braking force = $50\% \times$

$$486\text{kN} = 243\text{kN}$$

This force will need to be transmitted from the roof slab to the walls; the continuity reinforcement between the two members will need to resist a shearing force of $243 / 20.6 = 12\text{kN/m}$

Partial Factors γ_F (From NA to BS EN 1990:2002+A1:2005)

For SLS factors see Clause NA.2.3.9

For EQU factors see Table NA.A2.4(A)

For STR/GEO Comb1 factors see Table NA.A2.4(B)

For STR/GEO Comb2 factors see Table NA.A2.4(C)

PD 6694-1:2011 Clause 10.2.2: Model Factor for negative arching effects for structures not founded on hard material $= \gamma_{Sd;ec,max} = 1.15 + 0.35(H_c - 8)/3$ but not less than 1.15

$$\gamma_{Sd;ec,max} = 1.15 + 0.35(1.0 - 8)/3 = 0.33 < 1.15 \therefore = \mathbf{1.15}$$

Allow for deviation of road construction thickness to UK NA 1991-1-1 Table NA.1 cl.5.2.3(3) = +55% or -40%.

Unfavourable	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
Self weight of structure & backfill $\gamma_{G;sup}$	1.0	1.05	1.35	1.0
Superimposed permanent load $\gamma_{G;sup}$	1.0	1.05	1.2	1.0
Road traffic action on box $\gamma_{Q;sup}$	1.0	1.35	1.35	1.15
Thermal Actions $\gamma_{Q;sup}$	1.0	1.55	1.55	1.3
Material Factor to ϕ' γ_M	1.0	1.1	1.0	1.25
Vertical and horizontal water pressures $\gamma_{G;sup}$	0.0	1.0	1.0	1.0

Favourable	SLS	EQU	STR/GEO Comb1)	STR/GEO Comb2
Self weight of structure & backfill $\gamma_{G;inf}$	1.0	0.95	0.95	1.0
Superimposed permanent load $\gamma_{G;inf}$	1.0	0.95	0.95	1.0

Analysis

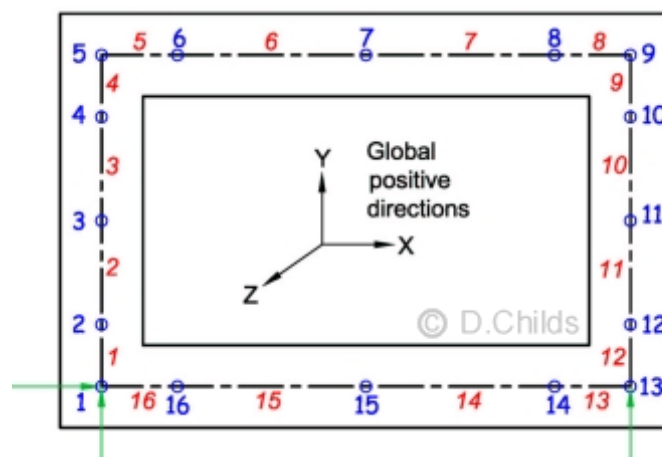
An efficient method of analysis is to use a plane frame model; a [spreadsheet](#) is available from this website.

Using C32/40 concrete then the short-term modulus of elasticity $E_{cm} = 22[(f_{ck} + 8) / 10]^{0.3} = 33\text{kN/mm}^2$.

For analysing permanent loads then it is usual to use the long-term value for E_{cm} which can be taken as half the short-term value $= 33 / 2 = 17\text{kN/mm}^2$.

Shear is considered at a distance d (effective depth of cross-section) away from the support. Nodes (2, 4, 6, 8, 10, 12, 14 & 16) are positioned on the plane frame model at these points to determine the appropriate shear value. An estimate of the concrete cover and reinforcement bar size will be required to determine a value for d . For the example assume member thickness = 300mm, bar size = 20mm and cover = 60mm

Then $d = 300 - 60 - 10 = 230\text{mm}$.



Artificial supports need to be provided to the model at nodes 1 and 13. Node 1 is fixed in the X & Y directions and free to rotate. Node 13 is fixed in the Y direction only. When loading (which includes bearing pressure) is applied to the model then the net reactions at the supports should be zero.

It is usual to analyse the structure using nominal loads for each action and then combine the design values using the appropriate partial factors.

D1) Self Weight Analysis

Assuming 300mm thick for walls, roof and floor then load $= 0.3 \times 25 = 7.5\text{kN/m}$

Total load for concrete $= 76.5\text{kN}$

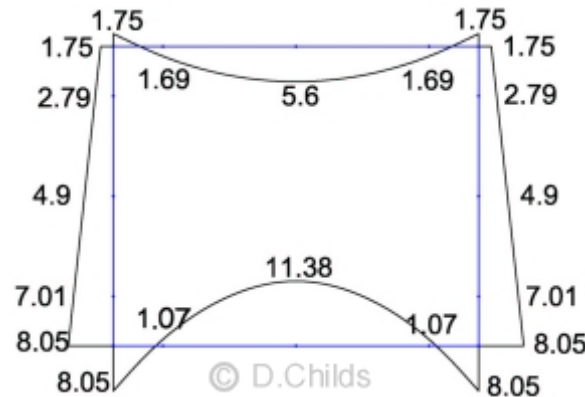
Bearing pressure $= 76.5 / 3.1 = 24.7\text{kN/m}$

The plane frame analysis idealises the structure as a series of beam elements positioned at the centre of the members. Consequently the model dimensions of the culvert are 2.8m wide $\times$ 2.3m high. Equilibrium will be maintained in the model by determining the bearing pressure on the model base length of 2.8m. The pressure applied to the model $= 76.5 / 2.8 = 27.32\text{kN/m}$

The bearing pressure can therefore be applied (upwards) as a UDL of 27.32kN/m to the floor members.

The bending moment **results under self weight** are shown. Values are displayed on the tension face of the member.

Note: If the correct loading and bearing pressure has been applied to the model then the model reactions at nodes 1 and 13 should be zero.

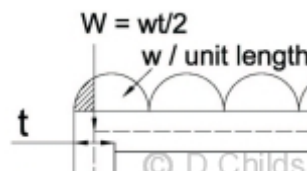


(D1) Self Weight Moments

D2) Road Construction Analysis

200mm thick surfacing weight = 4.8kN/m applied as UDL to roof members.

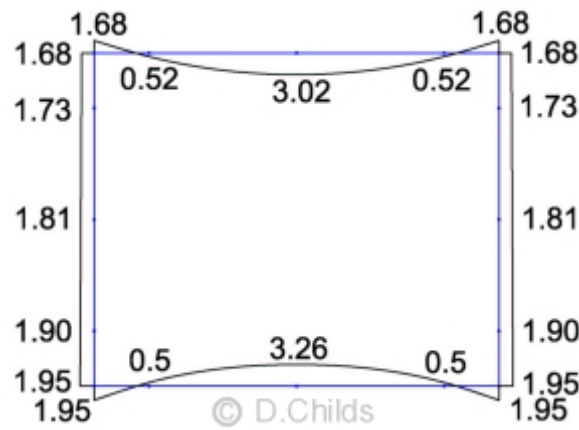
However the loading is applied to the outside dimensions (3.1m wide × 2.6m high). A correction can be made by applying point loads at the corners of the model to represent the load outside the centre-line of the corner member.



Point load corrections applied at each corner = $0.15 \times 4.8 = 0.72\text{kN}$

Bearing pressure = 4.8kN/m. UDL applied to floor members = $4.8 \times 3.1 / 2.8 = 5.31\text{kN/m}^2$.

The bending moment **results under road construction weight** are shown. Values are displayed on the tension face of the member.

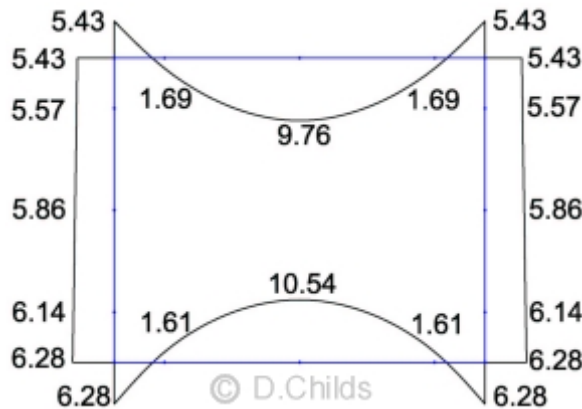


(D2) Road Construction Moments

D3) Fill Over Box Analysis

800mm thick fill weight = 15.5kN/m applied as UDL to roof members.

The analysis is similar to the road construction so the results can be multiplied by $15.5 / 4.8 = 3.23$ to obtain fill over results.

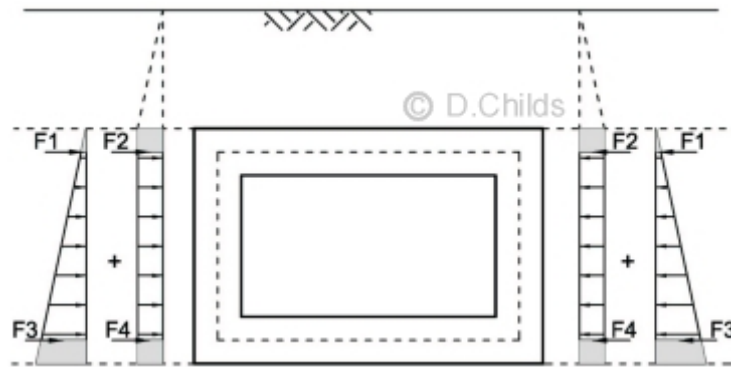


(D3) Fill Over Box Moments

D4, D5 & D6) Earth Pressure on Box Walls Analysis

The earth pressure to the side walls comprises pressure from the backfill between foundation level up to top-of-roof level, and surcharge from the soil/carriageway construction above roof level.

These pressures are modelled as trapezoidal loads for the backfill and a UDL for the surcharge, together with the point load corrections representing the pressure beyond the centre-line of the roof and floor slabs.



Earth Pressures on Walls Moments

The vertical pressure at top-of-roof level from the carriageway and fill = $4.8 + 15.5 = 20.3$ kN/m (per metre width of box).

Surcharge pressure on walls = $20.3 \times K_d$ kN/m.

Point load corrections $F2 = F4 = 0.15 \times 20.3 \times K_d = 3.05 \times K_d$ kN.

Use $K_d = 0.6$ (Table B.1 SLS K_{max}) then:

UDL = $20.3 \times 0.6 = 12.18$ kN/m

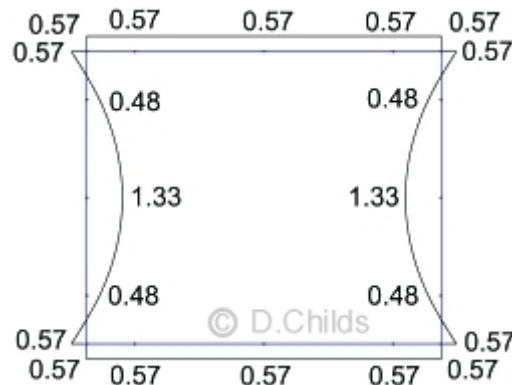
$F2 = F4 = 3.05 \times 0.6 = 1.83$ kN

The bending moment [results under earth surcharge pressures](#) on walls for road construction and fill can be obtained separately by factoring the results.

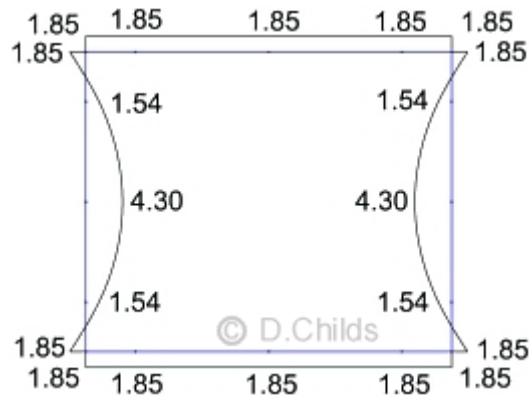
Road construction surcharge moment factor = $4.8 / 20.3 = 0.236$

Fill surcharge moment factor = $15.5 / 20.3 = 0.764$

Values are displayed on the tension face of the member.



(D4) Moments from Road Construction Surcharge Pressures



(D5) Moments from Fill Surcharge Pressures

Use $K_d = 0.6$ (Table B.1 SLS K_{max}) then:

The backfill pressure varies from 0 at the top-of-roof level down to $K_d \times \gamma_{b'fill} \times H = 0.6 \times 19 \times 2.6 = 29.64 \text{ kN/m}$ at foundation level.

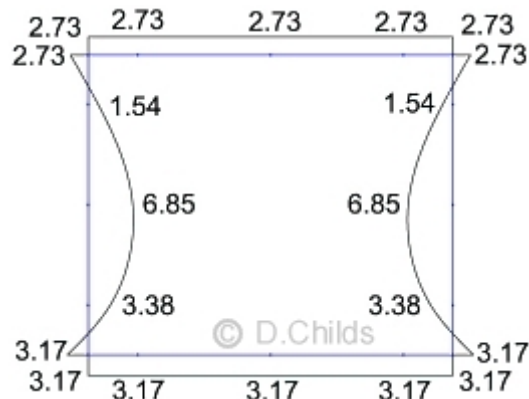
Pressure at centre-line of roof slab = $0.6 \times 19 \times 0.15 = 1.71 \text{ kN/m}$.

Pressure at centre-line of floor slab = $0.6 \times 19 \times 2.45 = 27.93 \text{ kN/m}$.

Point load correction $F1 = 0.5 \times 1.71 \times 0.15 = 0.13 \text{ kN}$.

Point load correction $F3 = 0.5 \times (27.93 + 29.64) \times 0.15 = 4.32 \text{ kN}$.

The bending moment [results under earth backfill pressures](#) on walls are shown. Values are displayed on the tension face of the member.

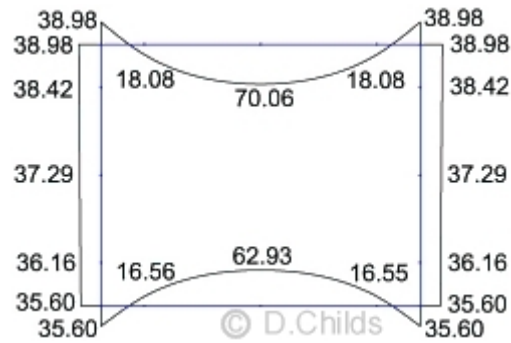


(D6) Earth Backfill Pressures on Walls Moments

D7) Vertical Traffic Loading Analysis

The moving load spreadsheet shows that LM1 traffic is critical. When moving loads are analysed an envelope of moments (together with shears and axial loads) are produced for the maximum load effect at each node as the vehicle crosses the structure.

The bending moment [results from the spreadsheet](#) are shown. Values are displayed on the tension face of the member.



(D7) LM1 Envelope of Maximum Moments

D8) Horizontal Traffic Surcharge Analysis

Horizontal Line Load $F = 55K_d \text{ kN}$ (see above).

UDL for LM1 = $20K_d R = 20K_d \text{ (kN/m}^2\text{)}$

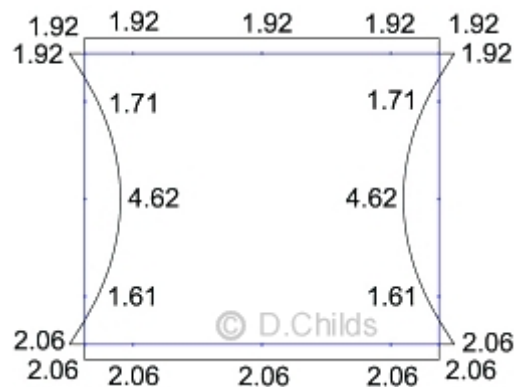
Use $K_d = 0.5$ (Table B.1 SLS K_0) then:

$$F = 55 \times 0.5 = 27.5 \text{ kN}$$

$$\text{UDL} = 20 \times 0.5 = 10 \text{ kN/m}$$

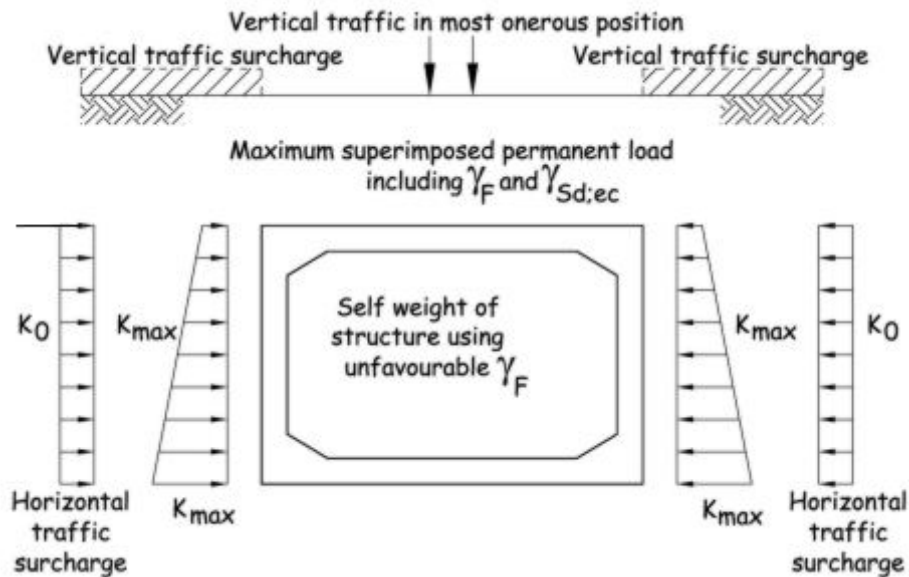
$$\text{Point load correction } F_2 = F_4 = 10 \times 0.15 = 1.5 \text{ kN.}$$

The bending moment [results from the spreadsheet](#) are shown. Values are displayed on the tension face of the member.



(D8) LM1 Surcharge Pressures on Walls Moments

Load Case Table B.1



The bending moments for actions shown in [Table B.1](#) are obtained by applying the appropriate partial factors to the moment values obtained from the plane frame analysis results.

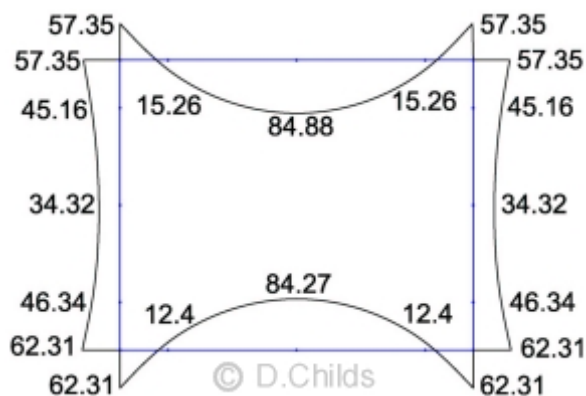
The maximum load effects in the members will be obtained with no water inside the culvert.

A simple spreadsheet was set up to combine the factored moments for the Limit States SLS, EQU, STR/GEO Combination 1 and STR/GEO Combination 2. The [results](#) are shown below.

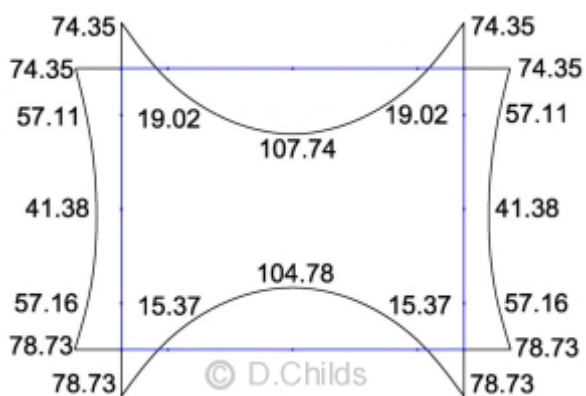
Diagram	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
D1	1.0	1.05	1.35	1.0
D2	$1.0 \times 1.15 \times 1.55 = 1.78$	$1.05 \times 1.15 \times 1.55 = 1.87$	$1.2 \times 1.15 \times 1.55 = 2.14$	$1.0 \times 1.15 \times 1.55 = 1.78$
D3	$1.0 \times 1.15 = 1.15$	$1.05 \times 1.15 = 1.21$	$1.2 \times 1.15 = 1.38$	$1.0 \times 1.15 = 1.15$
D4	$1.0 \times 1.55 \times 0.6 / 0.6 = 1.55$	$1.05 \times 1.55 \times 0.77 / 0.6 = 2.09$	$1.35 \times 1.55 \times 0.72 / 0.6 = 2.51$	$1.0 \times 1.55 \times 0.84 / 0.6 = 2.17$
D5	$1.0 \times 0.6 / 0.6 = 1.0$	$1.05 \times 0.77 / 0.6 = 1.35$	$1.35 \times 0.72 / 0.6 = 1.62$	$1.0 \times 0.84 / 0.6 = 1.40$
D6	$1.0 \times 0.6 / 0.6 = 1.0$	$1.05 \times 0.77 / 0.6 = 1.35$	$1.35 \times 0.72 / 0.6 = 1.62$	$1.0 \times 0.84 / 0.6 = 1.40$

D7	1.0	1.35	1.35	1.15
D8	$1.0 \times 0.5 / 0.5$ = 1.0	$1.35 \times 0.54 / 0.5$ = 1.46	$1.35 \times 0.5 / 0.5$ = 1.35	$1.15 \times 0.58 / 0.5$ = 1.33

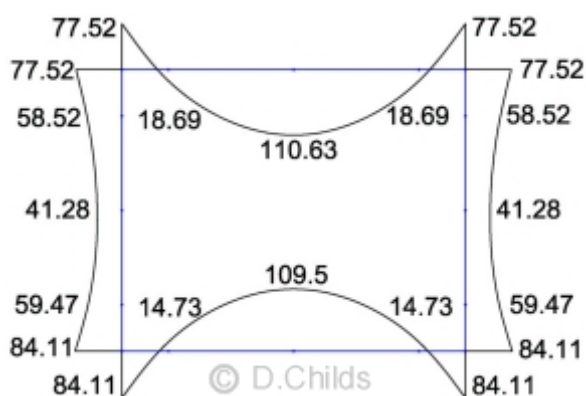
Limit State Results for Table B.1 Loading:



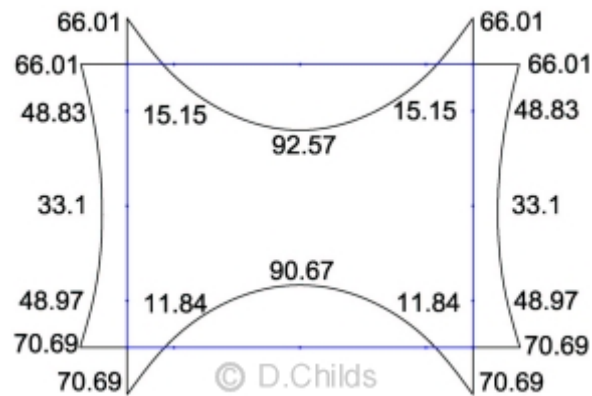
SLS Moments



EQU Moments

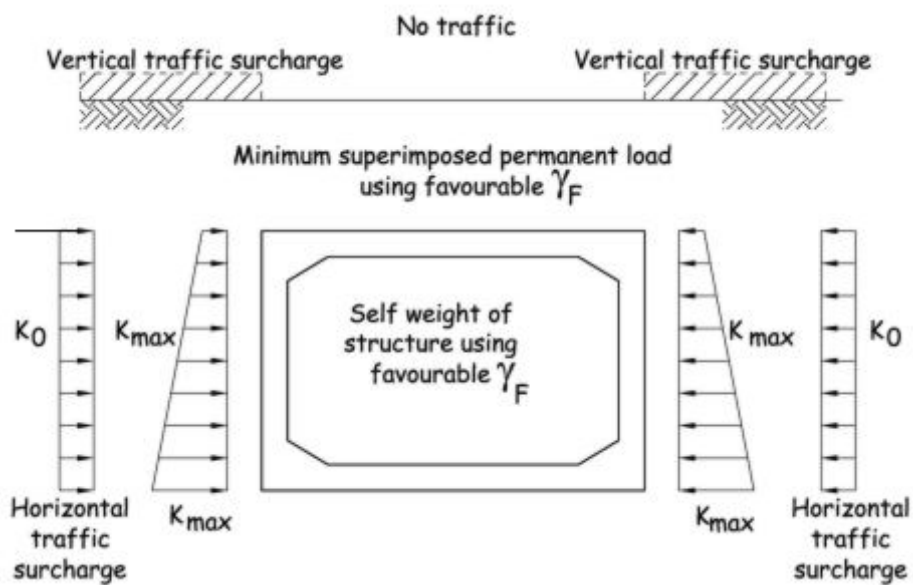


STR/GEO Comb.1 Moments



STR/GEO Comb.2 Moments

Load Case Table B.2



The bending moments for actions shown in [Table B.2](#) are obtained by applying the appropriate partial factors to the moment values obtained from the plane frame analysis results.

The maximum load effects in the members will be obtained with no water inside the culvert.

A simple spreadsheet was set up to combine the factored moments for the Limit States SLS, EQU, STR/GEO Combination 1 and STR/GEO Combination 2. The [results](#) are shown below.

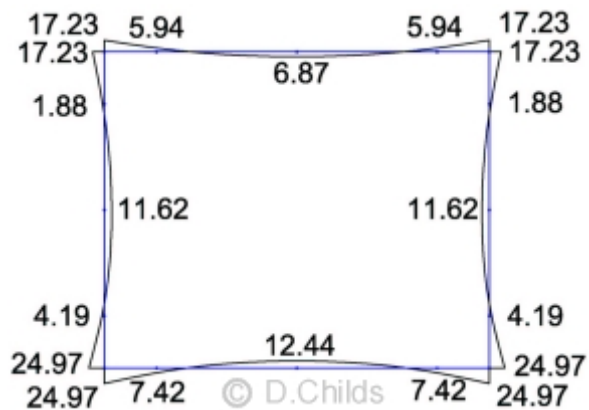
Diagram	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
D1	1.0	0.95	0.95	1.0

D2	$1.0 \times 0.6 = 0.6$	$0.95 \times 0.6 = 0.57$	$0.95 \times 0.6 = 0.57$	$1.0 \times 0.6 = 0.6$
D3	1.0	0.95	0.95	1.0
D4	$1.0 \times 0.6 \times 0.6 / 0.6 = 0.6$	$1.05 \times 0.6 \times 0.77 / 0.6 = 0.81$	$1.35 \times 0.6 \times 0.72 / 0.6 = 0.97$	$1.0 \times 0.6 \times 0.84 / 0.6 = 0.84$
D5	$1.0 \times 0.6 / 0.6 = 1.0$	$1.05 \times 0.77 / 0.6 = 1.35$	$1.35 \times 0.72 / 0.6 = 1.62$	$1.0 \times 0.84 / 0.6 = 1.40$
D6	$1.0 \times 0.6 / 0.6 = 1.0$	$1.05 \times 0.77 / 0.6 = 1.35$	$1.35 \times 0.72 / 0.6 = 1.62$	$1.0 \times 0.84 / 0.6 = 1.40$
D7	0	0	0	0
D8	$1.0 \times 0.5 / 0.5 = 1.0$	$1.35 \times 0.54 / 0.5 = 1.46$	$1.35 \times 0.5 / 0.5 = 1.35$	$1.15 \times 0.58 / 0.5 = 1.33$

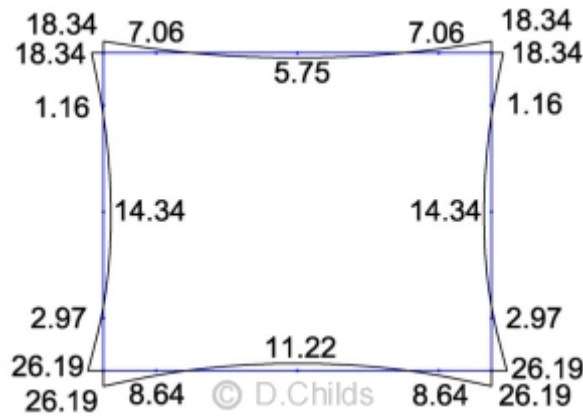
Limit State Results for Table B.2 Loading:



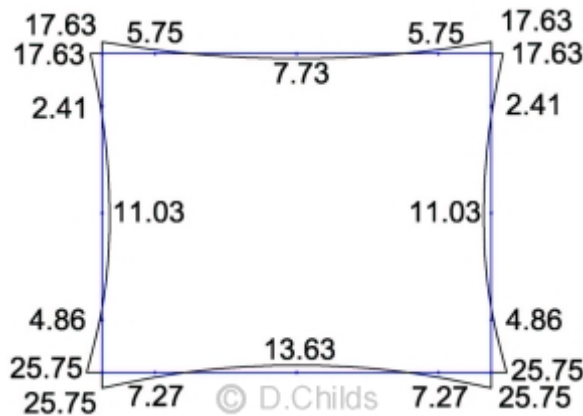
SLS Moments



EQU Moments

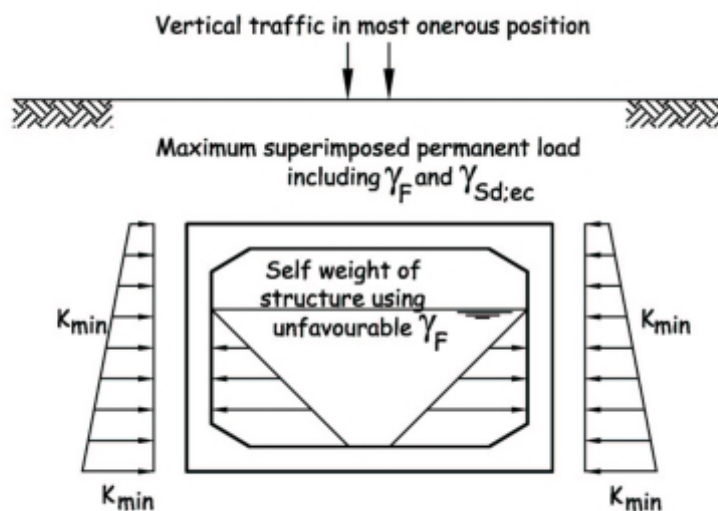


STR/GEO Comb.1 Moments



STR/GEO Comb.2 Moments

Load Case Table B.3



The bending moments for actions shown in [Table B.3](#) are obtained by applying the appropriate partial factors to the moment values obtained from the plane frame analysis

results.

Two conditions will be analysed:

1. with water inside the culvert
2. with no water inside the culvert

Depth of water = 1.8m so pressure on wall at base = $1.8 \times 10 = 18\text{kN/m}$

Pressure from water on pane frame model floor slab = $18 \times 2.5 / 2.8 = 16.07\text{kN/m}$

Bearing pressure from water = $18 \times 2.5 / 3.1 = 14.52\text{kN/m}$

Triangular loads are applied to the walls of the plane frame model, the UDLs of 16.07kN/m on the floor slab representing the water pressure and bearing pressure will cancel out.

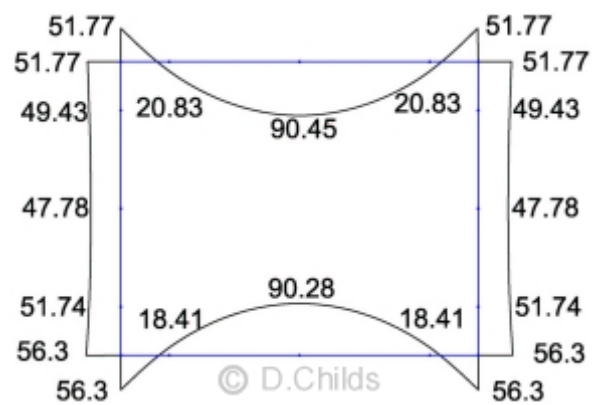
The [results from the plane frame analysis](#) are included in the combination with the appropriate partial factors. Note that the water pressure on the walls has been applied to members 1 to 3 and 10 to 12 for convenience; this has slightly overestimated the depth (1.92m instead of 1.8m). However the load effects are not great and can be considered acceptable.

The maximum load effects in the members will be obtained by comparing the results. A simple spreadsheet was set up to combine the factored moments for the Limit States SLS, EQU, STR/GEO Combination 1 and STR/GEO Combination 2. The [results](#) are shown below.

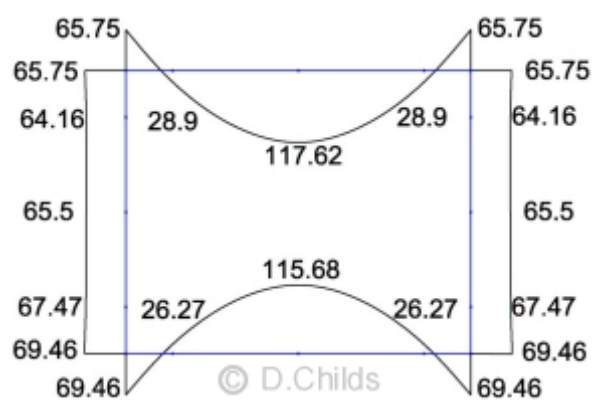
Diagram	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
D1	1.0	1.05	1.35	1.0
D2	$1.0 \times 1.15 \times$ 1.55 = 1.78	$1.05 \times 1.15 \times$ 1.55 = 1.87	$1.2 \times 1.15 \times$ 1.55 = 2.14	$1.0 \times 1.15 \times$ 1.55 = 1.78
D3	$1.0 \times 1.15 =$ 1.15	$1.05 \times 1.15 =$ 1.21	$1.2 \times 1.15 =$ 1.38	$1.0 \times 1.15 =$ 1.15
D4	$1.0 \times 1.55 \times$ $0.2 / 0.6 =$ 0.52	$0.95 \times 1.55 \times$ $0.18 / 0.6 =$ 0.44	$0.95 \times 1.55 \times$ $0.2 / 0.6 =$ 0.49	$1.0 \times 1.55 \times$ $0.16 / 0.6 =$ 0.41
D5	$1.0 \times 0.2 / 0.6$ = 0.33	$0.95 \times 0.18 /$ 0.6 = 0.29	$0.95 \times 0.2 /$ 0.6 = 0.32	$1.0 \times 0.16 /$ 0.6 = 0.27
D6	$1.0 \times 0.2 / 0.6$ = 0.33	$0.95 \times 0.18 /$ 0.6 = 0.29	$0.95 \times 0.2 /$ 0.6 = 0.32	$1.0 \times 0.16 /$ 0.6 = 0.27
D7	1.0	1.35	1.35	1.15

D8	0	0	0	0
Water	0	1.0	1.0	1.0

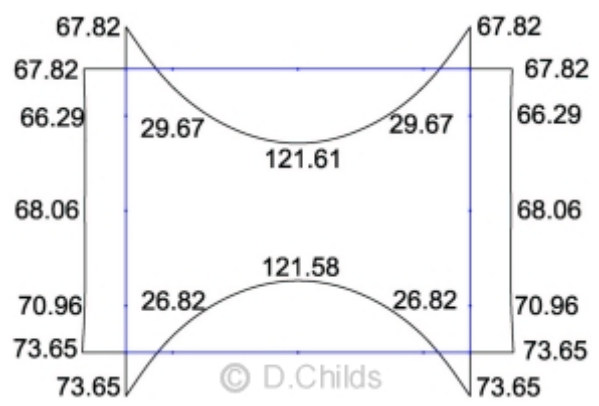
Limit State Results for Table B.3 Loading:



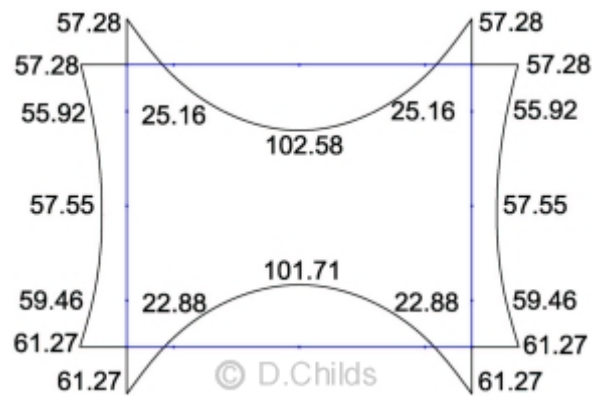
SLS Moments



EQU Moments

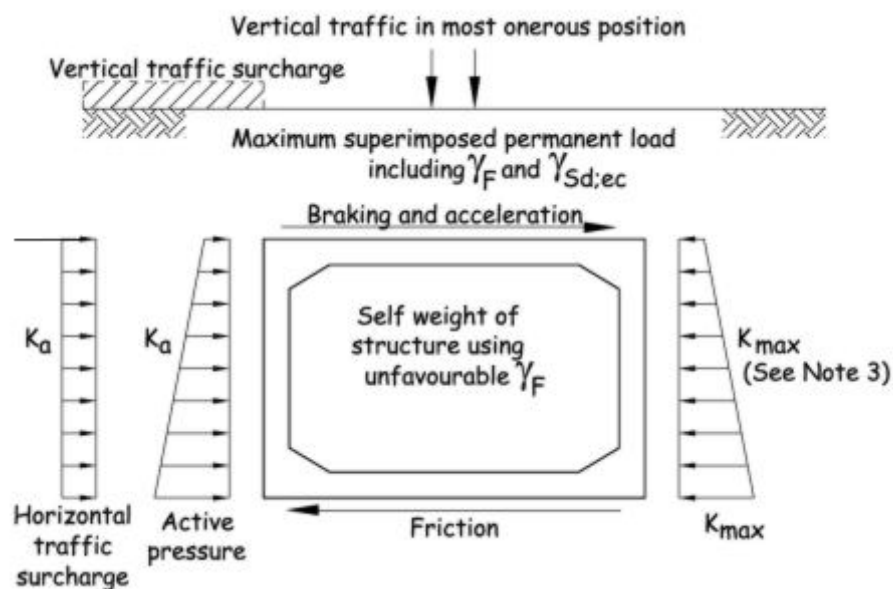


STR/GEO Comb.1 Moments



STR/GEO Comb.2 Moments

Load Case Table B.4



Stability needs to be considered by assessing the resistance to sliding and overturning together with the bearing pressure to PD 6694-1:2011 Clauses 10.3.2 and 10.3.3. Load effects in the members will need to be considered if $K_{\max}$ has to be increased above the values given in PD 6694-1:2011 Table B.4. Partial Factors can be determined as before.

Consider horizontal forces to check sliding:

For drained conditions design resistance to sliding $R_d = V'_d \tan \delta_d$ where δ = structure-ground interface friction angle = 32°

Applying γ_M to $\tan \delta$ we get $\delta_d = \tan^{-1}(\tan \delta / \gamma_M)$

SLS: $\delta_d = \tan^{-1}(\tan 32 / 1.0) = 32^\circ$

EQU: $\delta_d = \tan^{-1}(\tan 32 / 1.1) = 29.6^\circ$

$$\text{ULS Comb.1: } \delta_d = \tan^{-1}(\tan 32 / 1.0) = 32^\circ$$

$$\text{ULS Comb.2: } \delta_d = \tan^{-1}(\tan 32 / 1.25) = 26.6^\circ$$

Sliding at SLS:

$\gamma_F = 1.0$ for all elements.

Maximum surcharge from road construction = $1.55 \times 4.8 = 7.44 \text{ kN/m}$

Maximum surcharge from fill above roof level = 15.5 kN/m

Active force from surcharge = $0.33 \times (7.44 + 15.5) \times 2.6 = 19.68 \text{ kN}$

Passive force from surcharge = $0.6 \times (7.44 + 15.5) \times 2.6 = 35.79 \text{ kN}$

Backfill pressure at foundation level = $19 \times 2.6 = 49.4 \text{ kN/m}^2$

Active force from backfill up to roof = $0.33 \times 49.4 \times 2.6 / 2 = 21.19 \text{ kN}$

Passive force from backfill up to roof = $0.6 \times 49.4 \times 2.6 / 2 = 38.53 \text{ kN}$

Active force from traffic surcharge line load = $0.33 \times 55 = 18.15 \text{ kN}$

Active force from LM1 traffic UDL surcharge = $0.33 \times 20 \times 2.6 = 17.16 \text{ kN}$

Active force from LM3 traffic UDL surcharge = $0.33 \times 30 \times 2.6 = 25.74 \text{ kN}$

Braking and acceleration force for LM1 traffic = 15 kN

Braking and acceleration force for LM3 traffic = 24 kN

Total active force with LM3 traffic = $19.68 + 21.19 + 18.15 + 25.74 + 24 = 108.76 \text{ kN}$

Total passive force = $35.79 + 38.53 = 74.32 \text{ kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $1.15 \times 1.55 \times 4.8 \times 3.1 = 26.53 \text{ kN}$

Fill on roof = $1.15 \times 15.5 \times 3.1 = 55.26 \text{ kN}$

Self weight of concrete = 76.5 kN

LM1 traffic UDL = $5.5 \times 3.1 = 17.05 \text{ kN}$

LM1 traffic Tandem System = $85.0 \times 1.555 \times 2 = 264 \text{ kN}$

LM3 traffic = $40.8 \times (2 \times 1.505 + 0.7) = 151 \text{ kN}$

By inspection LM3 traffic critical (maximum horizontal with minimum vertical loading).

$V'_d = 26.53 + 55.26 + 76.5 + 151 = 309.29 \text{ kN}$

Maximum $R_d = 309.29 \times \tan 32^\circ = 193 \text{ kN}$

Friction required = $108.76 - 74.32 = 34.44 < 193 \therefore$ sliding can be resisted using

$K_{\max} = 0.6 \therefore \text{OK.}$

Sliding at EQU:

Active force from surcharge = $0.44 \times 1.05 \times (7.44 + 15.5) \times 2.6 = 27.56 \text{ kN}$

Passive force from surcharge = $0.6 \times 1.05 \times (7.44 + 15.5) \times 2.6 = 37.58 \text{ kN}$

Active force from backfill up to roof = $0.44 \times 1.05 \times 49.4 \times 2.6 / 2 = 29.67 \text{ kN}$

Passive force from backfill up to roof = $0.6 \times 1.05 \times 49.4 \times 2.6 / 2 = 40.46 \text{ kN}$

Active force from traffic surcharge line load = $0.37 \times 1.35 \times 55 = 27.47 \text{ kN}$

Active force from LM3 traffic UDL surcharge = $0.37 \times 1.35 \times 30 \times 2.6 = 38.96 \text{ kN}$

Braking and acceleration force for LM3 traffic = $1.35 \times 24 = 32.4\text{kN}$

Total active force with LM3 traffic = $27.56 + 29.67 + 27.47 + 38.96 + 32.4 = 156.06\text{kN}$

Total passive force = $37.58 + 40.46 = 78.04\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $1.05 \times 26.53 = 27.86\text{kN}$

Fill on roof = $1.05 \times 55.26 = 58.02\text{kN}$

Self weight of concrete = $1.05 \times 76.5 = 80.33\text{kN}$

LM3 traffic = $1.35 \times 151 = 204\text{kN}$

$V'_d = 27.86 + 58.02 + 80.33 + 204 = 370\text{kN}$

Maximum $R_d = 370 \times \tan 29.6^\circ = 210\text{kN}$

Friction required = $156.06 - 78.04 = 78.02 < 210 \therefore$ sliding can be resisted using $K_{\max} = 0.6 \therefore$ OK.

Sliding at ULS comb.1:

Active force from surcharge = $0.40 \times 1.20 \times (7.44 + 15.5) \times 2.6 = 28.63\text{kN}$

Passive force from surcharge = $0.72 \times 1.20 \times (7.44 + 15.5) \times 2.6 = 51.53\text{kN}$

Active force from backfill up to roof = $0.40 \times 1.35 \times 49.4 \times 2.6 / 2 = 34.68\text{kN}$

Passive force from backfill up to roof = $0.72 \times 1.35 \times 49.4 \times 2.6 / 2 = 62.42\text{kN}$

Active force from traffic surcharge line load = $0.33 \times 1.35 \times 55 = 24.5\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.33 \times 1.35 \times 30 \times 2.6 = 34.75\text{kN}$

Braking and acceleration force for LM3 traffic = $1.35 \times 24 = 32.4\text{kN}$

Total active force with LM3 traffic = $28.63 + 34.68 + 24.5 + 34.75 + 32.4 = 154.96\text{kN}$

Total passive force = $51.53 + 62.42 = 113.95\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $1.20 \times 26.53 = 31.84\text{kN}$

Fill on roof = $1.20 \times 55.26 = 66.31\text{kN}$

Self weight of concrete = $1.35 \times 76.5 = 103.28\text{kN}$

LM3 traffic = $1.35 \times 151 = 204\text{kN}$

$V'_d = 31.84 + 66.31 + 103.28 + 204 = 405\text{kN}$

Maximum $R_d = 405 \times \tan 32^\circ = 253\text{kN}$

Friction required = $154.96 - 113.95 = 41.01 < 253 \therefore$ sliding can be resisted using $K_{\max} = 0.72 \therefore$ OK.

Sliding at ULS comb.2

Active force from surcharge = $0.49 \times 1.0 \times (7.44 + 15.5) \times 2.6 = 29.23\text{kN}$

Passive force from surcharge = $0.84 \times 1.0 \times (7.44 + 15.5) \times 2.6 = 50.1\text{kN}$

Active force from backfill up to roof = $0.49 \times 1.0 \times 49.4 \times 2.6 / 2 = 31.47\text{kN}$

Passive force from backfill up to roof = $0.84 \times 1.0 \times 49.4 \times 2.6 / 2 = 53.94\text{kN}$

Active force from traffic surcharge line load = $0.41 \times 1.15 \times 55 = 25.93\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.41 \times 1.15 \times 30 \times 2.6 = 36.78\text{kN}$

Braking and acceleration force for LM3 traffic = $1.15 \times 24 = 27.6\text{kN}$

Total active force with LM3 traffic = $29.23 + 31.47 + 25.93 + 36.78 + 27.6 = 151.01\text{kN}$

Total passive force = $50.1 + 53.94 = 104.04\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $1.0 \times 26.53 = 26.53\text{kN}$

Fill on roof = $1.0 \times 55.26 = 55.26\text{kN}$

Self weight of concrete = $1.0 \times 76.5 = 76.5\text{kN}$

LM3 traffic = $1.15 \times 151 = 174\text{kN}$

$V'_d = 26.53 + 55.26 + 76.5 + 174 = 332\text{kN}$

Maximum $R_d = 332 \times \tan 26.6^\circ = 166\text{kN}$

Friction required = $151.01 - 104.04 = 46.97 < 166 \therefore$ sliding can be resisted using

$K_{\max} = 0.84 \therefore$ OK.

Overturning:

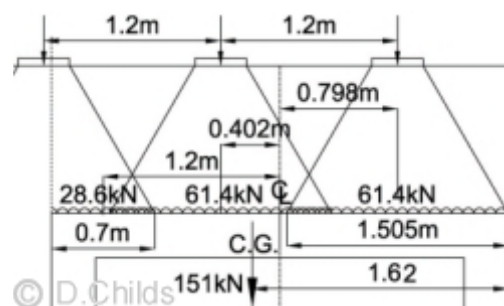
Stability against overturning is found by considering the eccentricity of loads about the centre of the foundation.

PD 6694-1 cl. 5.2.2 says it is prudent to ensure that no uplift occurs anywhere under the foundations at SLS (eccentricity to be kept within the middle third of the base length).

BS EN 1997-1 cl. 6.5.4 says that special precautions shall be taken where the eccentricity of loading exceeds 1/3 of the width of a rectangular footing at ULS (eccentricity to be kept within middle two thirds of base length).

Eccentricity of Traffic Loading:

Maximum vertical traffic loading from trailer axles:



LM3 patch loading on 1m width of roof

Distance to centre of gravity of patch loads = $[40.8 \times 1.505 \times (1.505 + 1.2) + 40.8 \times 0.7 \times (3.1 - 0.35)] / 151 = 1.62\text{m}$

As the SV196 trailer has at least 4 axles at 1.2m centres then the patch loads can be moved to form the mirror image of the diagram shown, so the 1.62m can be measured from the front edge or back edge of the culvert.

The eccentricity of the load is therefore $\pm (1.62 - 3.1 / 2) = \pm 0.07\text{m}$.

Consider SLS bearing pressure under base:

Total vertical load = $26.53 + 55.26 + 151 + 76.5 = 309.29\text{kN}$

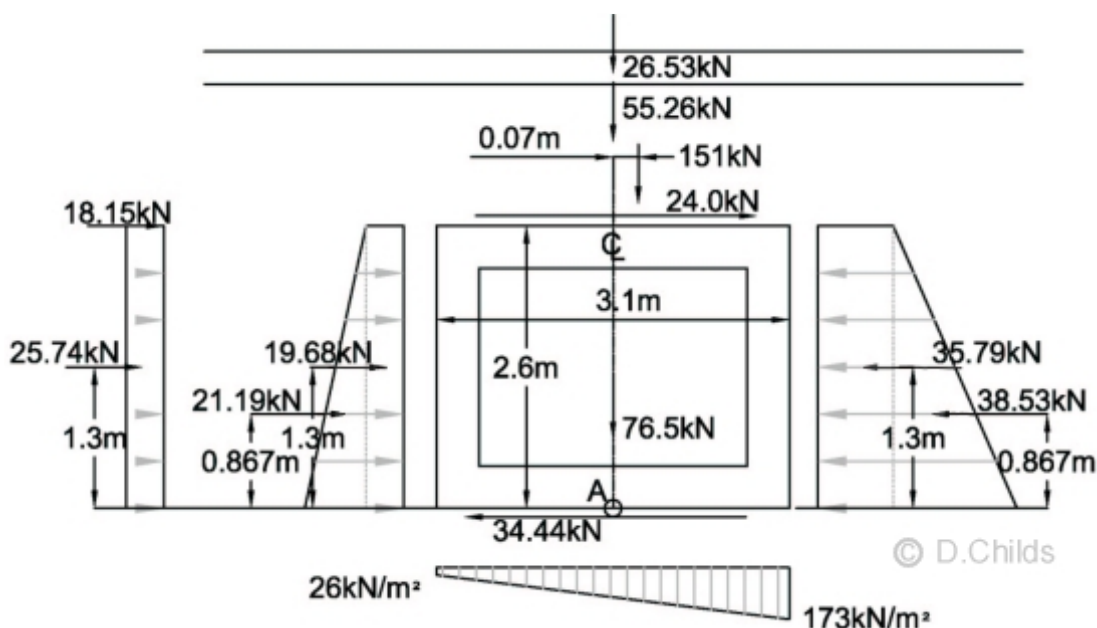
Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (18.15 + 24) + 1.3 \times (25.74 + 19.68 - 35.79) + 0.867 \times (21.19 - 38.53) + 151 \times 0.07 = 117.6\text{kNm}$

Eccentricity $e = M / V = 117.6 / 309.29 = 0.38\text{m}$.

No uplift will occur if $e < \text{base length} / 6 = 3.1 / 6 = 0.517 > 0.38\text{m} \therefore \text{OK}$.

The bearing pressure diagram will be trapezoidal at SLS so pressures = $V/A \pm M / Z = 309.29 / (1.0 \times 3.1) \pm 117.6 / (1.0 \times 3.1^2 / 6) = 99.77 \pm 73.42 = 173 \text{ and } 26 \text{ kN/m}^2$.



Bearing Pressure under Base at SLS

Check for eccentricity of loads at ULS:

For **EQU Limit State**:

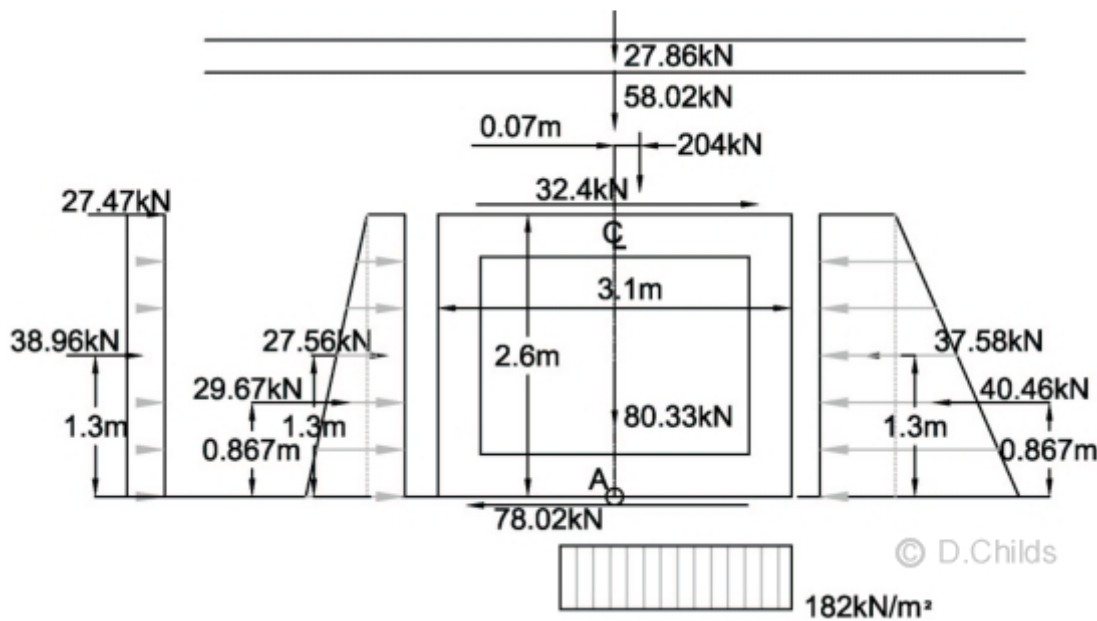
Total vertical load = $27.86 + 58.02 + 151 \times 1.35 + 80.33 = 370.06\text{kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (27.47 + 32.4) + 1.3 \times (38.96 + 27.56 - 37.58) + 0.867 \times (29.67 - 40.46) + 204 \times 0.07 = 198.21\text{kNm}$

Eccentricity $e = M / V = 198.21 / 370.06 = 0.536\text{m}$.

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.03 > 0.536\text{m} \therefore \text{OK}$.
 The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 370.06 / (3.1 - 2 \times 0.536) = 182 \text{ kN/m}^2$.



Bearing Pressure under Base at EQU

For **STR/GEO (Comb 1) Limit State:**

Total vertical load = $31.84 + 66.31 + 151 \times 1.35 + 103.28 = 405.28\text{kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (24.5 + 32.4) + 1.3 \times (34.75 + 28.63 - 51.53) + 0.867 \times (34.68 - 62.42) + 204 \times 0.07 = 153.57\text{kNm}$

Eccentricity $e = M / V = 153.57 / 405.28 = 0.379\text{m}$.

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.03 > 0.379\text{m} \therefore \text{OK}$.

The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 405.28 / (3.1 - 2 \times 0.379) = 173 \text{ kN/m}^2$.

For **STR/GEO (Comb 2) Limit State:**

Total vertical load = $26.53 + 55.26 + 151 \times 1.15 + 76.5 = 331.94\text{kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (25.93 + 27.6) + 1.3 \times (36.78 + 29.23 - 50.1) + 0.867 \times (31.47 - 53.94) + 174 \times 0.07 = 152.56\text{kNm}$

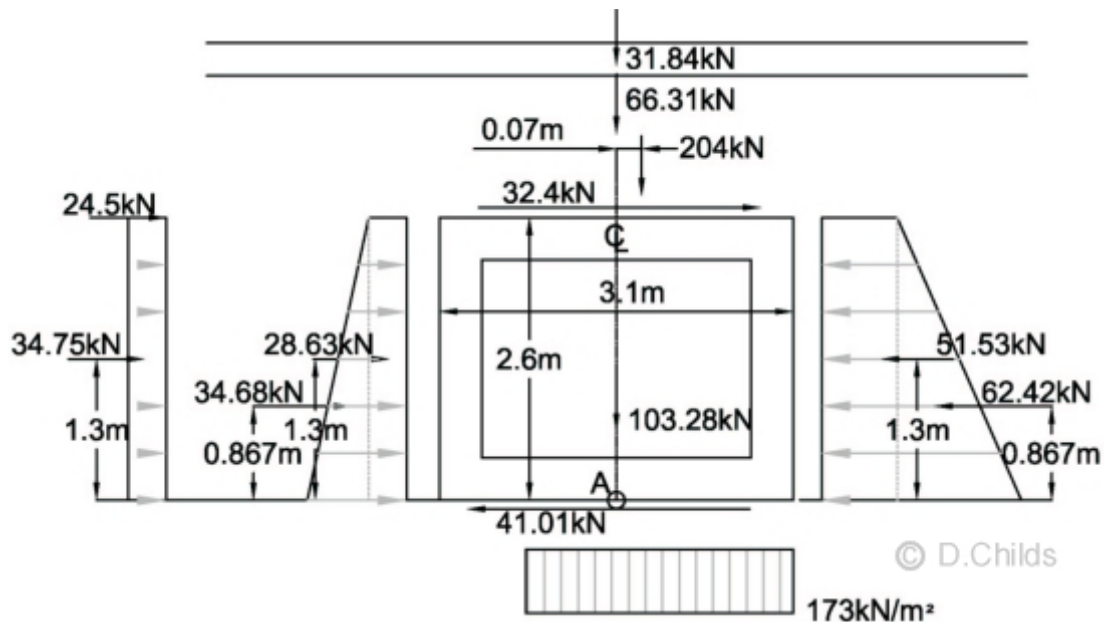
Eccentricity $e = M / V = 152.56 / 331.94 = 0.460\text{m}$.

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.03 > 0.460\text{m} \therefore \text{OK}$.

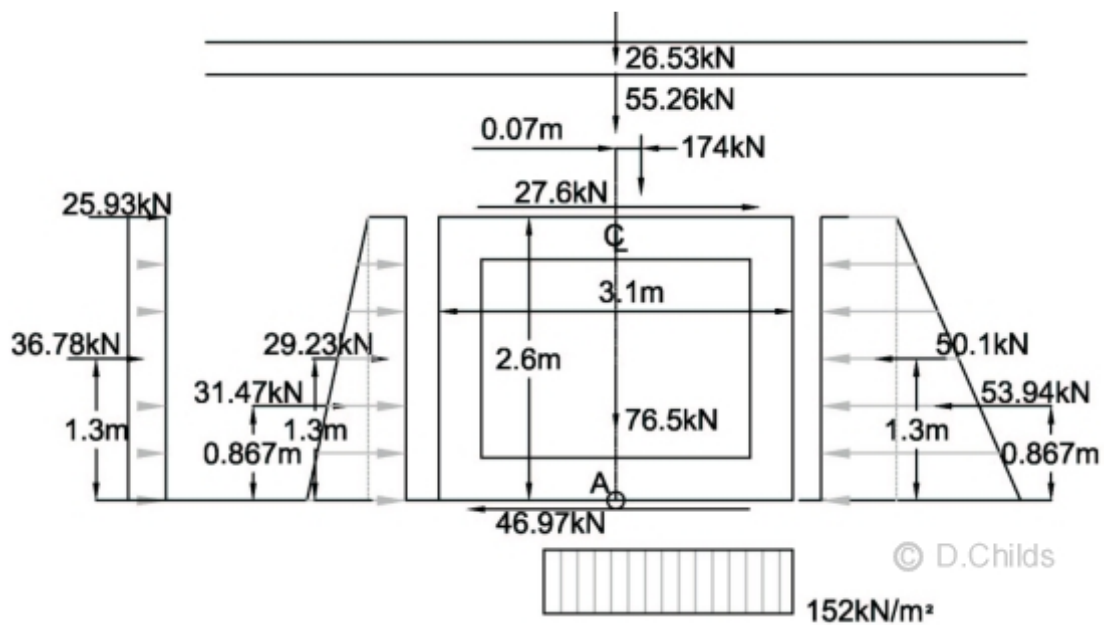
The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 331.94 / (3.1 - 2 \times 0.46) = 152 \text{ kN/m}^2$.

Check Bearing Resistance from BS EN 1997-1 Clause D.4

Equation (D.2): $R/A' = c'N_c b_{cs} i_{cs} + q'N_q b_{qs} i_{qs} + 0.5\gamma_d B'N_\gamma b_\gamma s_\gamma i_\gamma$



Bearing Pressure under Base at STR/GEO (Comb 1)



Bearing Pressure under Base at STR/GEO (Comb 2)

$c' = 0$ (cohesionless soil)

$q' = 1.5 \times 19 = 28.5 \text{ kN/m}^2$ (overburden pressure)

$\gamma' = 19 \text{ kN/m}^3$ (weight density of foundation material)

$N_q = e^{\pi \tan \phi'_d} \tan^2(45 + \phi'_d/2)$

$N_\gamma = 2 (N_q - 1) \tan \phi'$, where $\delta \geq \phi' / 2$ (rough base)

$b_q = b_\gamma = (1 - \alpha \tan \phi'_d)$

Inclination $\alpha = 0$

$s_q = 1 + (B' / L') \sin \phi'_d$

$s_\gamma = 1 - 0.3(B' / L')$

$m = m_B = [2 + (B' / L')] / [1 + (B' / L')] \text{ when } H \text{ acts in the direction of } B'$

$$i_q = [1 - H / (V + A'c'_d \cot \phi'_d)]^m$$

$$i_\gamma = [1 - H / (V + A'c'_d \cot \phi'_d)]^{m+1}$$

From Figure D.1:

$L' = 20.6\text{m}$ (transverse width of foundation)

$B = 3.1\text{m}$ $B' = B - 2e_B$

For 1m strip design $A' = 1.0 \times B' = B'$

	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
e_B	0.38	0.536	0.379	0.46
$A' \text{ \& } B'$	$3.1 - 2 \times 0.38 = 2.224$	$3.1 - 2 \times 0.536 = 2.028$	$3.1 - 2 \times 0.379 = 2.342$	$3.1 - 2 \times 0.46 = 2.18$
$\gamma_d = \gamma' \times \gamma_{G;\text{inf}}$	$19.0 \times 1.0 = 19.0$	$19.0 \times 0.95 = 18.05$	$19.0 \times 0.95 = 18.05$	$19.0 \times 1.0 = 19.0$
ϕ'_d	$\tan^{-1}(\tan 32 / 1.0) = 32^\circ$	$\tan^{-1}(\tan 32 / 1.1) = 29.6^\circ$	$\tan^{-1}(\tan 32 / 1.0) = 32^\circ$	$\tan^{-1}(\tan 32 / 1.25) = 26.6^\circ$
N_q	23.18	17.59	23.18	12.64
N_γ	27.72	18.85	27.72	11.66
$b_q \text{ \& } b_\gamma$	1.0	1.0	1.0	1.0
s_q	1.057	1.049	1.06	1.047
s_γ	0.968	0.97	0.966	0.968
m	1.903	1.91	1.898	1.904
Net H_d	$108.76 - 74.32 = 34.44$	$156.06 - 78.04 = 78.02$	$154.96 - 113.95 = 41.01$	$151.01 - 101.04 = 49.97$
V_d	309.29	370.06	405.28	331.94
i_q	0.799	0.636	0.817	0.733

i_γ	0.71	0.502	0.734	0.623
$q'N_q b_q s_q i_q$	558	334	572	276
$0.5\gamma_d B' N_\gamma b_\gamma s_\gamma i_\gamma$	426	171	422	151
R / A'	984	505	994	427
V_d / A'	$309.29 / 2.224 = 139$	$370.06 / 2.028 = 182$	$405.28 / 2.342 = 173$	$331.94 / 2.18 = 152$
R / V_d	7.08	2.77	5.75	2.81

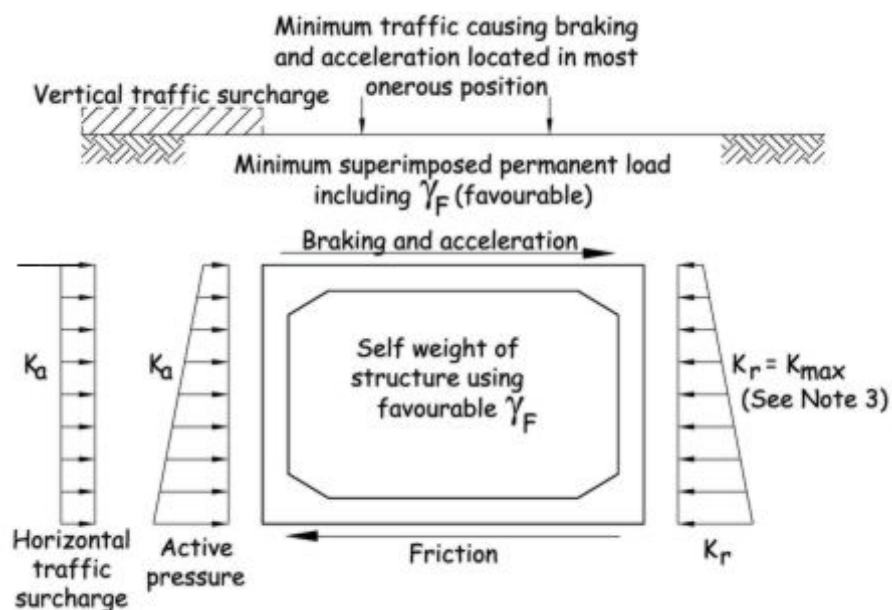
PD 6694-1:2011 Clause 5.2.2 says the serviceability limit state for settlement can be verified by ensuring that a "sufficiently low fraction of the ground strength is mobilized". This requirement can be deemed to be satisfied if the maximum pressure under a foundation at SLS does not exceed one third of the design resistance R/A' .

$$1/3 (R / A') = 984 / 3 = 328 \text{ kN/m}^2$$

Maximum bearing pressure at SLS = $173 \text{ kN/m}^2 < 328 \therefore$ settlement requirement can be deemed to be satisfied.

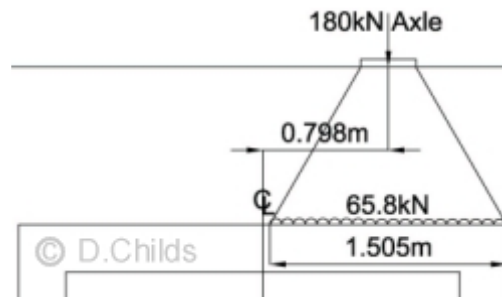
Minimum Bearing Resistance Factor = $2.77 > 1.0 \therefore$ O.K.

Load Case Table B.5



This is similar to Table B.4 loading except vertical loads are treated as favourable. Load effects in the members will need to be considered if $K_{\max}$ has to be increased above the values given in PD 6694-1:2011 Table B.5.

Consider tractor axle in worst position



65.8kN LM3 patch loading on 1m width of roof at $e = 0.798\text{m}$

Consider the maximum friction that can be developed between the 180kN axle and the roof of the structure:

Maximum vertical load from LM3 on a unit width of box = 65.8kN.

Vertical load from fill above culvert = $(15.5 + 4.8 \times 0.6) \times 3.1 = 56.98\text{kN}$

Assuming a coefficient of friction of $\tan 30^\circ$ then:

Maximum friction that can be generated on a metre width between the earth and the roof is $(65.8 + 56.98)\tan 30^\circ = 71\text{kN}$

Total braking force generated by SV196 vehicle = $25 + 2 \times 45 + 9 \times 41.25 = 486\text{kN}$

Using the in-plane rigidity of the roof and walls as proven above then braking/acceleration force = $486 / 20.6 = 23.6\text{kN/m} < 71\text{kN} \therefore$ use 23.6kN/m

Sliding at SLS:

$\gamma_F = 1.0$ for all elements.

Minimum surcharge from road construction = $0.6 \times 4.8 = 2.88\text{kN/m}$

Minimum surcharge from fill above roof level = 15.5kN/m

Active force from surcharge = $0.33 \times (2.88 + 15.5) \times 2.6 = 15.77\text{kN}$

Passive force from surcharge = $0.6 \times (2.88 + 15.5) \times 2.6 = 28.67\text{kN}$

Backfill pressure at foundation level = $19 \times 2.6 = 49.4\text{kN/m}^2$

Active force from backfill up to roof = $0.33 \times 49.4 \times 2.6 / 2 = 21.19\text{kN}$

Passive force from backfill up to roof = $0.6 \times 49.4 \times 2.6 / 2 = 38.53\text{kN}$

Active force from traffic surcharge line load = $0.33 \times 55 = 18.15\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.33 \times 30 \times 2.6 = 25.74\text{kN}$

Braking and acceleration force for LM3 traffic = 23.6kN

Total active force with LM3 traffic = $15.77 + 21.19 + 18.15 + 25.74 + 23.6 = 104.45\text{kN}$

Total passive force = $28.67 + 38.53 = 67.2\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

$$\text{Road construction} = 0.6 \times 4.8 \times 3.1 = 8.93\text{kN}$$

$$\text{Fill on roof} = 15.5 \times 3.1 = 48.05\text{kN}$$

$$\text{Self weight of concrete} = 76.5\text{kN}$$

$$\text{LM3 traffic} = 65.8\text{kN}$$

$$V'_d = 8.93 + 48.05 + 76.5 + 65.8 = 199.28\text{kN}$$

$$\text{Maximum } R_d = 199.28 \times \tan 32^\circ = 125\text{kN}$$

$$\text{Friction required} = 104.85 - 67.2 = 38 < 125 \therefore \text{sliding can be resisted using } K_{\max} = 0.6 \therefore \text{OK.}$$

Sliding at EQU:

$$\text{Active force from surcharge} = 0.44 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 19.98\text{kN}$$

$$\text{Passive force from surcharge} = 0.6 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 27.24\text{kN}$$

$$\text{Active force from backfill up to roof} = 0.44 \times 0.95 \times 49.4 \times 2.6 / 2 = 26.84\text{kN}$$

$$\text{Passive force from backfill up to roof} = 0.6 \times 0.95 \times 49.4 \times 2.6 / 2 = 36.61\text{kN}$$

$$\text{Active force from traffic surcharge line load} = 0.37 \times 1.35 \times 55 = 27.47\text{kN}$$

$$\text{Active force from LM3 traffic UDL surcharge} = 0.37 \times 1.35 \times 30 \times 2.6 = 38.96\text{kN}$$

$$\text{Braking and acceleration force for LM3 traffic} = 1.35 \times 23.6 = 31.9\text{kN}$$

$$\text{Total active force with LM3 traffic} = 19.98 + 26.84 + 27.47 + 38.96 + 31.9 = 145.15\text{kN}$$

$$\text{Total passive force} = 27.24 + 36.61 = 63.85\text{kN}$$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

$$\text{Road construction} = 0.95 \times 8.93 = 8.48\text{kN}$$

$$\text{Fill on roof} = 0.95 \times 48.05 = 45.65\text{kN}$$

$$\text{Self weight of concrete} = 0.95 \times 76.5 = 72.68\text{kN}$$

$$\text{LM3 traffic} = 1.35 \times 65.8 = 88.83\text{kN}$$

$$V'_d = 8.48 + 45.65 + 72.68 + 88.83 = 215.64\text{kN}$$

$$\text{Maximum } R_d = 215.64 \times \tan 29.6^\circ = 123\text{kN}$$

$$\text{Friction required} = 145.15 - 63.85 = 81 < 123 \therefore \text{sliding can be resisted using } K_{\max} = 0.6 \therefore \text{OK.}$$

Sliding at ULS comb.1:

$$\text{Active force from surcharge} = 0.40 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 18.16\text{kN}$$

$$\text{Passive force from surcharge} = 0.72 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 32.69\text{kN}$$

$$\text{Active force from backfill up to roof} = 0.40 \times 0.95 \times 49.4 \times 2.6 / 2 = 24.40\text{kN}$$

$$\text{Passive force from backfill up to roof} = 0.72 \times 0.95 \times 49.4 \times 2.6 / 2 = 43.93\text{kN}$$

$$\text{Active force from traffic surcharge line load} = 0.33 \times 1.35 \times 55 = 24.5\text{kN}$$

$$\text{Active force from LM3 traffic UDL surcharge} = 0.33 \times 1.35 \times 30 \times 2.6 = 34.75\text{kN}$$

Braking and acceleration force for LM3 traffic = $1.35 \times 23.6 = 31.86\text{kN}$
Total active force with LM3 traffic = $18.16 + 24.4 + 24.5 + 34.75 + 31.86 = 133.67\text{kN}$

Total passive force = $32.69 + 43.93 = 76.62\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $0.95 \times 8.93 = 8.48\text{kN}$

Fill on roof = $0.95 \times 48.05 = 45.65\text{kN}$

Self weight of concrete = $0.95 \times 76.5 = 72.68\text{kN}$

LM3 traffic = $1.35 \times 65.8 = 88.83\text{kN}$

$V'_d = 8.48 + 45.65 + 72.68 + 88.83 = 215.64\text{kN}$

Maximum $R_d = 215.64 \times \tan 32^\circ = 135\text{kN}$

Friction required = $133.67 - 76.62 = 57 < 135 \therefore$ sliding can be resisted using $K_{\max} = 0.72 \therefore$ OK.

Sliding at ULS comb.2

Active force from surcharge = $0.49 \times 1.0 \times (2.88 + 15.5) \times 2.6 = 23.42\text{kN}$

Passive force from surcharge = $0.84 \times 1.0 \times (2.88 + 15.5) \times 2.6 = 40.14\text{kN}$

Active force from backfill up to roof = $0.49 \times 1.0 \times 49.4 \times 2.6 / 2 = 31.47\text{kN}$

Passive force from backfill up to roof = $0.84 \times 1.0 \times 49.4 \times 2.6 / 2 = 53.94\text{kN}$

Active force from traffic surcharge line load = $0.41 \times 1.15 \times 55 = 25.93\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.41 \times 1.15 \times 30 \times 2.6 = 36.78\text{kN}$

Braking and acceleration force for LM3 traffic = $1.15 \times 23.6 = 27.14\text{kN}$

Total active force with LM3 traffic = $23.42 + 31.47 + 25.93 + 36.78 + 27.14 = 144.7\text{kN}$

Total passive force = $40.14 + 53.94 = 94.08\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $1.0 \times 8.93 = 8.93\text{kN}$

Fill on roof = $1.0 \times 48.05 = 48.05\text{kN}$

Self weight of concrete = $1.0 \times 76.5 = 76.5\text{kN}$

LM3 traffic = $1.15 \times 65.8 = 75.67\text{kN}$

$V'_d = 8.93 + 48.05 + 76.5 + 75.67 = 209.15\text{kN}$

Maximum $R_d = 209.15 \times \tan 26.6^\circ = 105\text{kN}$

Friction required = $144.7 - 94.08 = 51 < 105 \therefore$ sliding can be resisted using $K_{\max} = 0.84 \therefore$ OK.

Overturning:

Stability against overturning is found by considering the eccentricity of loads about the

centre of the foundation.

Consider SLS bearing pressure under base:

Total vertical load = $8.93 + 48.05 + 65.8 + 76.5 = 199.28\text{kN}$

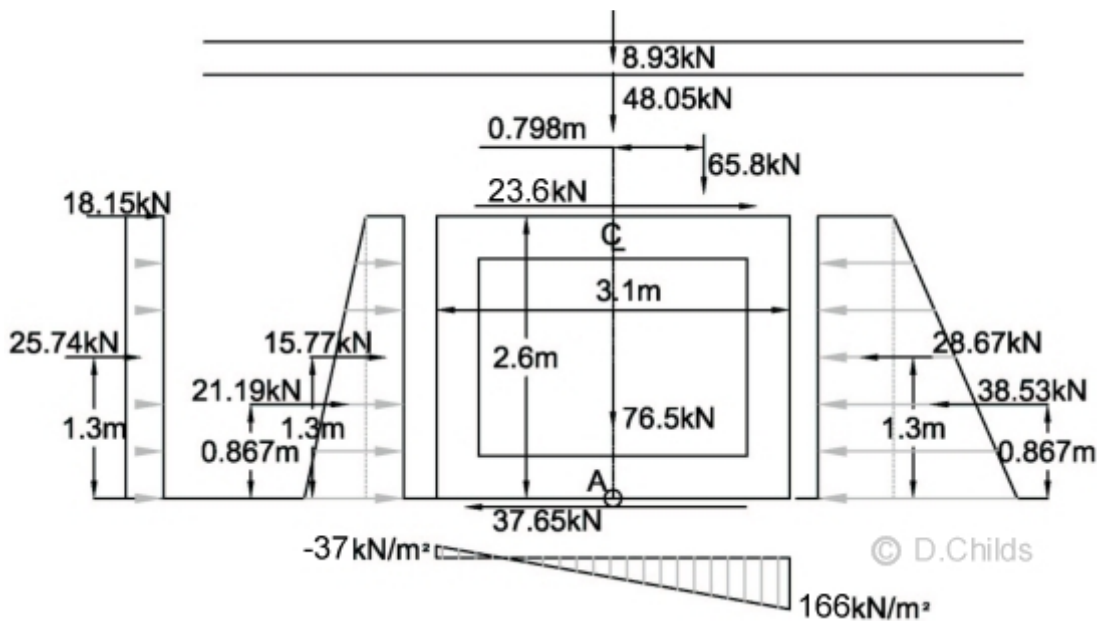
Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (18.15 + 23.6) + 1.3 \times (25.74 + 15.77 - 28.67) + 0.867 \times (21.19 - 38.53) + 65.8 \times 0.798 = 162.7\text{kNm}$

Eccentricity $e = M / V = 162.7 / 199.28 = 0.816\text{m}$.

No uplift will occur if $e < \text{base length} / 6 = 3.1 / 6 = 0.517 < 0.816\text{m} \therefore \text{Fail}$.

SLS pressures = $V/A \pm M / Z = 199.28 / (1.0 \times 3.1) \pm 162.7 / (1.0 \times 3.1^2 / 6) = 64.28 \pm 101.58 = 166 \text{ and } -37 \text{ kN/m}^2$.



Bearing Pressure under Base at SLS with uplift

Note 3 in Table B.5 says $K_{\max}$ may be increased to maintain stability, but check movement.

Active moment = $2.6 \times (18.15 + 23.6) + 1.3 \times (25.74 + 15.77) + 0.867 \times 21.19 + 65.8 \times 0.798 = 233.37\text{kNm}$

Passive moment (with $K_{\max} = 0.6$) = $1.3 \times 28.67 + 0.867 \times 38.53 = 70.68\text{kNm}$

To avoid uplift $V/A - M/Z > 0$

So $M < V \times Z / A = 199.28 \times 3.1^2 / (6 \times 3.1) = 102.96\text{kNm}$

Passive moment required = $233.37 - 102.96 = 130.41\text{kNm}$

Maximum bearing pressure = $199.28 / 3.1 + 102.96 \times 6 / 3.1^2 = 129\text{kN/m}^2$

$K_{\max}$ required = $0.6 \times 130.41 / 70.68 = 1.107$

Revised passive surcharge force = $28.67 \times 1.107 / 0.6 = 52.91\text{kN}$

Revised passive backfill force = $38.53 \times 1.107 / 0.6 = 71.11\text{kN}$

Check for movement to develop $K_{\max} = 1.107$:

From BS EN 1997-1:2004 Figure C.2.1 (with ϕ' for 6N backfill = 35°) $K_p = 3.7$

Proportion of K_p required = $1.107 / 3.7 = 0.3 (< 0.5)$

PD 6694-1:2011 Clause 7.5 says movement (v/h) varies linearly from zero to v_2/h as K increases from K_0 to $0.5K_p$

$$0.5K_p = 3.7 / 2 = 1.85$$

Tables B.1 and B.2 quote K_0 as 0.5 at SLS

$$\text{Proportion of } v_2/h \text{ movement} = (1.107 - 0.5) / (1.85 - 0.5) = 0.450$$

From BS EN 1997-1:2004 Table C.2 for dense soil the range of v_2/h for rotational movement = 1.1% to 2.0%

$$\text{Maximum movement} = 2\% \times 0.45 \times 2.6\text{m} = 23\text{mm}$$

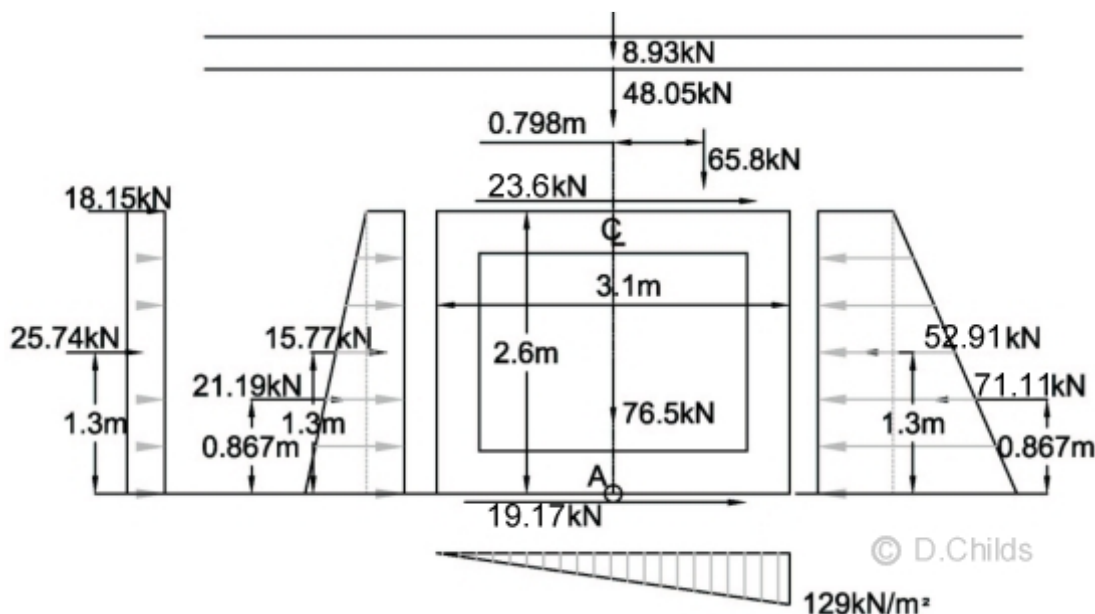
This movement would need to be checked for impact on adjacent services etc. and be accepted by the approving authority.

For lateral stability under increased passive force:

$$\text{Active horizontal force} = 15.77 + 21.19 + 18.15 + 25.74 + 23.6 = 104.85\text{kN}$$

$$\text{Passive horizontal force} = 52.91 + 71.11 = 124.02\text{kN}$$

For equilibrium friction on base = $124.02 - 104.85 = 19.17\text{kN}$ (in active direction) Although the passive force is greater than the active force it would still be prudent to check the structure for the increased loading.



Bearing Pressure under Base at SLS with increased $K_{\max}$.

Check for eccentricity of loads at ULS:

Consider EQU bearing pressure under base:

$$\text{Total vertical load} = 8.48 + 45.65 + 65.8 \times 1.35 + 72.68 = 215.64\text{kN}$$

Taking moments about the centre of the base (point A in diagram below):

$$\text{Moment} = 2.6 \times (27.47 + 31.9) + 1.3 \times (38.96 + 19.98 - 27.24) + 0.867 \times (26.84 - 36.61) + 88.83 \times 0.798 = 257.99\text{kNm}$$

$$\text{Eccentricity } e = M / V = 257.99 / 215.64 = 1.196\text{m.}$$

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.03 < 1.196\text{m} \therefore \text{Fail.}$

Moment required to limit e to $1.03\text{m} = 1.03 \times 215.64 = 222.11\text{kNm}$

Moment from passive forces (using $K_{\max} = 0.6$) $= 1.3 \times 27.24 + 0.867 \times 36.61 = 67.15$

$K_{\max}$ required to limit $e = 0.6 \times (67.15 + 257.99 - 222.11) / 67.15 = 0.92 (< 1.107 \text{ for SLS } \therefore \text{not critical})$

The bearing pressure diagram will be rectangular at ULS so pressure $= V / (L - 2e) = 215.64 / (3.1 - 2 \times 1.03) = 207.35\text{kN/m}^2$.

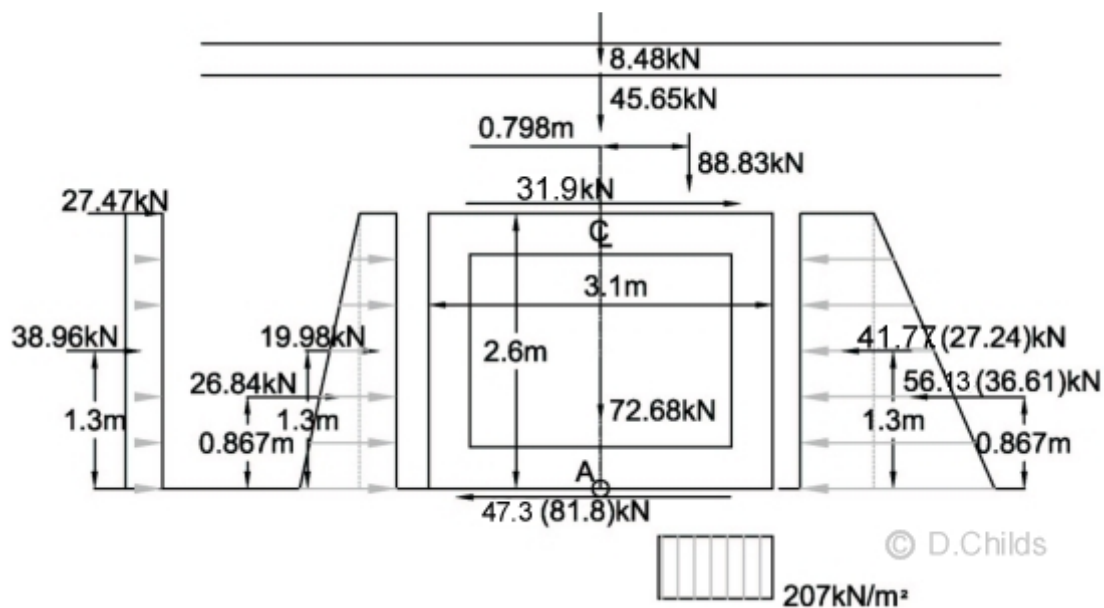
Passive force from surcharge $= 0.92 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 41.77\text{kN}$

Passive force from backfill up to roof $= 0.92 \times 0.95 \times 49.4 \times 2.6 / 2 = 56.13\text{kN}$

Total active force with LM3 traffic $= 19.98 + 26.84 + 27.47 + 38.96 + 31.9 = 145.15\text{kN}$

Total passive force $= 41.77 + 56.13 = 97.9\text{kN}$

Friction force on base to maintain stability $= 145.15 - 97.9 = 47.3\text{kN} < 123\text{kN} \therefore \text{O.K.}$



Bearing Pressure under Base at EQU using $K_{\max} = 0.92$ (Values in brackets using $K_{\max} = 0.6$)

Consider STR/GEO (Comb 1) bearing pressure under base:

Total vertical load $= 8.48 + 45.65 + 65.8 \times 1.35 + 72.68 = 215.64\text{kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment $= 2.6 \times (24.5 + 31.9) + 1.3 \times (34.75 + 18.16 - 32.69) + 0.867 \times (24.4 - 43.93) + 88.83 \times 0.798 = 226.88\text{kNm}$

Eccentricity $e = M / V = 226.88 / 215.64 = 1.05\text{m}$.

Check whether eccentricity exceeds base length / 3 $= 3.1 / 3 = 1.03 < 1.05\text{m} \therefore \text{Fail}$.

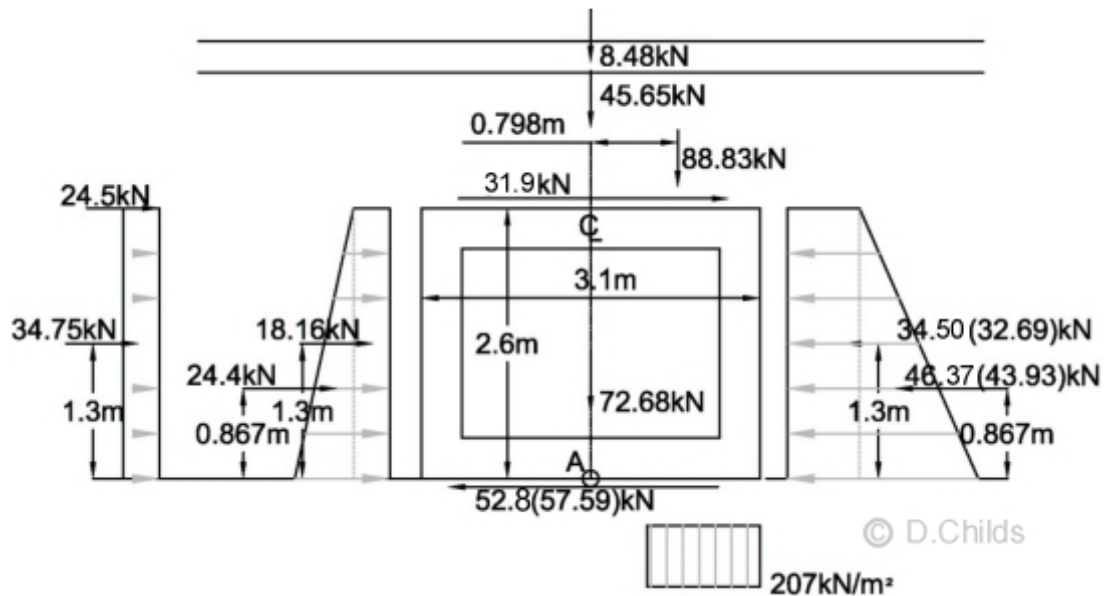
Moment required to limit e to $1.03\text{m} = 1.03 \times 215.64 = 222.11\text{kNm}$

Moment from passive forces (using $K_{\max} = 0.72$) $= 1.3 \times 32.69 + 0.867 \times 43.93 = 80.58$

$K_{\max}$ required to limit $e = 0.72 \times (80.58 + 226.88 - 222.11) / 80.58 = 0.76 (< 1.107 \text{ for SLS } \therefore \text{not critical})$

The bearing pressure diagram will be rectangular at ULS so pressure $= V / (L - 2e) = 215.64 / (3.1 - 2 \times 1.03) = 207\text{kN/m}^2$.

Passive force from surcharge = $0.76 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 34.5\text{kN}$
 Passive force from backfill up to roof = $0.76 \times 0.95 \times 49.4 \times 2.6 / 2 = 46.37\text{kN}$
 Total active force with LM3 traffic = $18.16 + 24.4 + 24.5 + 34.75 + 31.9 = 133.71\text{kN}$
 Total passive force = $34.5 + 46.37 = 80.87\text{kN}$
 Friction force on base to maintain stability = $133.71 - 80.87 = 52.8\text{kN} < 135\text{kN} \therefore \text{O.K.}$



Bearing Pressure under Base at STR/GEO (Comb 1) using $K_{\max} = 0.76$ (Values in brackets using $K_{\max} = 0.72$)

Consider STR/GEO (Comb 2) bearing pressure under base:

Total vertical load = $8.93 + 48.05 + 65.8 \times 1.15 + 76.5 = 209.15\text{kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (25.93 + 27.14) + 1.3 \times (36.78 + 23.42 - 40.14) + 0.867 \times (31.47 - 53.94) + 75.67 \times 0.798 = 204.96\text{kNm}$

Eccentricity $e = M / V = 204.96 / 209.15 = 0.98\text{m}$.

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.03 > 0.98\text{m} \therefore \text{OK}$.

The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 209.15 / (3.1 - 2 \times 0.98) = 183 \text{ kN/m}^2$.

Check Bearing Resistance from BS EN 1997-1 Clause D.4

Equation (D.2): $R/A' = c'N_c b_c s_{ci} c_i + q'N_q b_q s_{qi} q_i + 0.5\gamma_d B'N_\gamma b_\gamma s_{\gamma i} \gamma_i$

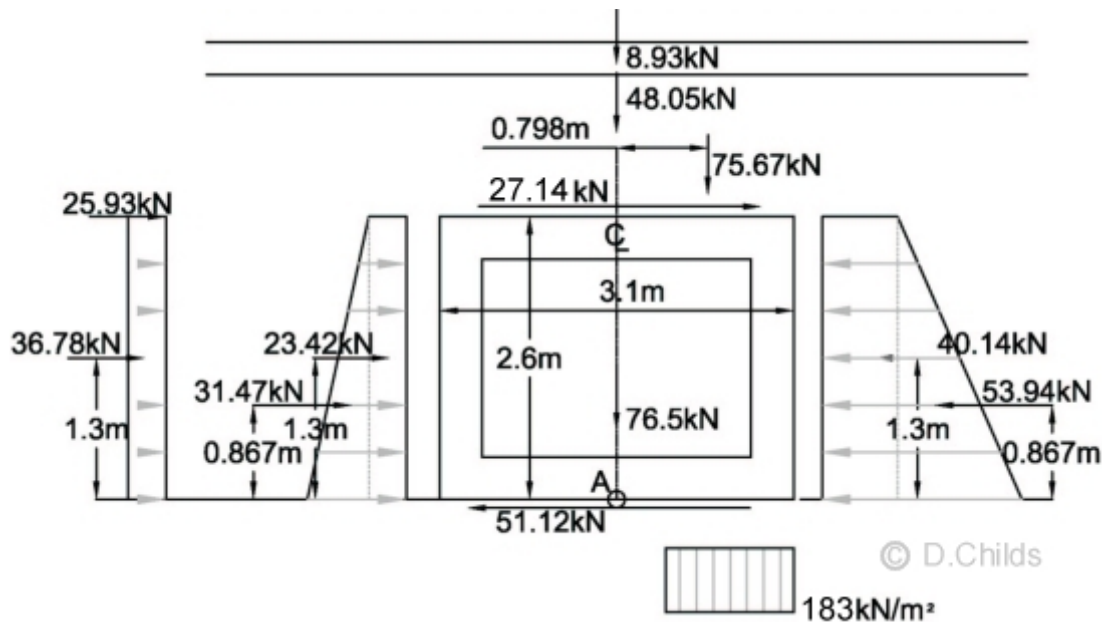
$c' = 0$ (cohesionless soil)

$q' = 1.5 \times 19 = 28.5 \text{ kN/m}^2$ (overburden pressure)

$\gamma' = 19 \text{ kN/m}^3$ (weight density of foundation material)

$N_q = e^{\pi \tan \phi' d} \tan^2(45 + \phi'_d / 2)$

$N_\gamma = 2 (N_q - 1) \tan \phi'$, where $\delta \geq \phi' / 2$ (rough base)



Bearing Pressure under Base at STR/GEO (Comb 2) using $K_{\max} = 0.84$

$$b_q = b_\gamma = (1 - \alpha \tan \phi'_d)$$

Inclination $\alpha = 0$

$$s_q = 1 + (B' / L') \sin \phi'_d$$

$$s_\gamma = 1 - 0.3(B' / L')$$

$m = m_B = [2 + (B' / L')] / [1 + (B' / L')]$ when H acts in the direction of B'

$$i_q = [1 - H / (V + A'c'_d \cot \phi'_d)]^m$$

$$i_\gamma = [1 - H / (V + A'c'_d \cot \phi'_d)]^{m+1}$$

From Figure D.1:

$L' = 20.6\text{m}$ (transverse width of foundation)

$B = 3.1\text{m}$ $B' = B - 2e_B$

For 1m strip design $A' = 1.0 \times B' = B'$

	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
e_B	0.517	1.03	1.03	0.98
A' & B'	$3.1 - 2 \times 0.517 = 2.066$	$3.1 - 2 \times 1.03 = 1.03$	$3.1 - 2 \times 1.03 = 1.03$	$3.1 - 2 \times 0.98 = 1.14$
$\gamma_d = \gamma' \times \gamma_{G;\inf}$	$19.0 \times 1.0 = 19.0$	$19.0 \times 0.95 = 18.05$	$19.0 \times 0.95 = 18.05$	$19.0 \times 1.0 = 19.0$
ϕ'_d	$\tan^{-1}(\tan 32 /$	$\tan^{-1}(\tan 32 /$	$\tan^{-1}(\tan 32 /$	$\tan^{-1}(\tan 32 /$

	1.0) = 32°	1.1) = 29.6°	1.0) = 32°	1.25) = 26.6°
N_q	23.18	17.59	23.18	12.64
N_γ	27.72	18.85	27.72	11.66
b_q & b_γ	1.0	1.0	1.0	1.0
s_q	1.053	1.025	1.027	1.024
s_γ	0.97	0.985	0.985	0.984
m	1.909	1.952	1.952	1.948
Net H_d	124.02 - 104.85 = 19.17	145.15 - 97.9 = 47.25	133.71 - 80.87 = 52.84	144.7 - 94.08 = 50.62
V_d	199.28	215.64	215.64	209.15
i_q	0.821	0.612	0.574	0.582
i_γ	0.74	0.476	0.432	0.441
$q'N_qb_qs_qi_q$	571	314	389	214
$0.5\gamma_dB'N_\gamma b_\gamma s_\gamma i_\gamma$	391	82	110	54
R / A'	962	396	500	269
V_d / A'	199.28 / 2.066 = 96	215.64 / 1.03 = 209	215.64 / 1.03 = 209	209.15 / 1.14 = 183
R / V_d	9.98	1.90	2.39	1.46

PD 6694-1:2011 Clause 5.2.2 says the serviceability limit state for settlement can be verified by ensuring that a "sufficiently low fraction of the ground strength is mobilized". This requirement can be deemed to be satisfied if the maximum pressure under a foundation at SLS does not exceed one third of the design resistance R/A' .

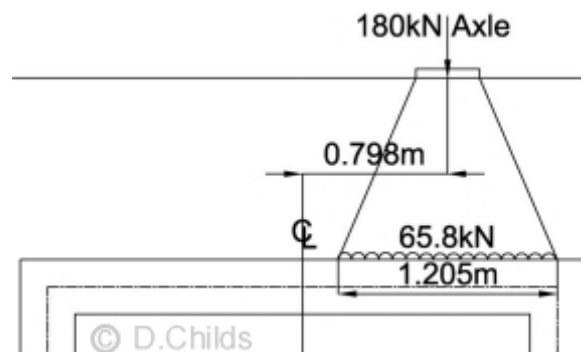
$$1/3 (R / A') = 962 / 3 = 321 \text{ kN/m}^2$$

Maximum bearing pressure at SLS = 129 kN/m² < 321 $\therefore$ settlement requirement can be

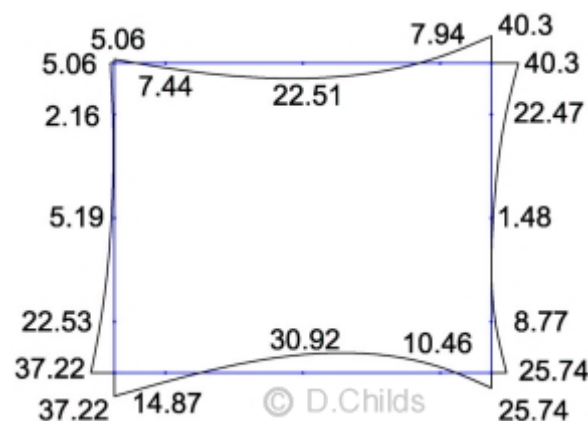
deemed to be satisfied.

Minimum Bearing Resistance Factor = $1.46 > 1.0 \therefore \text{O.K.}$

$K_{\max}$ had to be increased from the tabulated value for SLS, EQU and STR/GEO (comb.1) Limit States to maintain stability of the structure so we need to ensure that the structure is able to sustain the increased pressures. The plane frame analysis program is used to determine the additional load effects. Minor adjustments to the loading need to be made to minimise the magnitude of the artificial reactions at nodes 1 and 13.

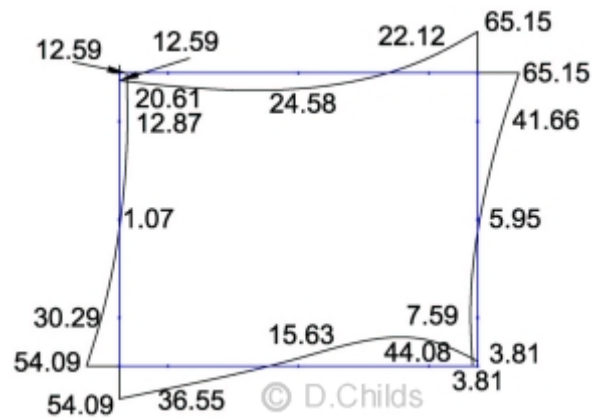


The length of the patch load for the LM3 vehicle will need to be shortened from 1.505m to $(1.505 - 2 \times 0.15) = 1.205\text{m}$ to keep the load within the limits of the model and also apply the same destabilising moment. Also the bearing pressures will need to be determined using the model base length (2.8m) instead of the structures base length (3.1m).



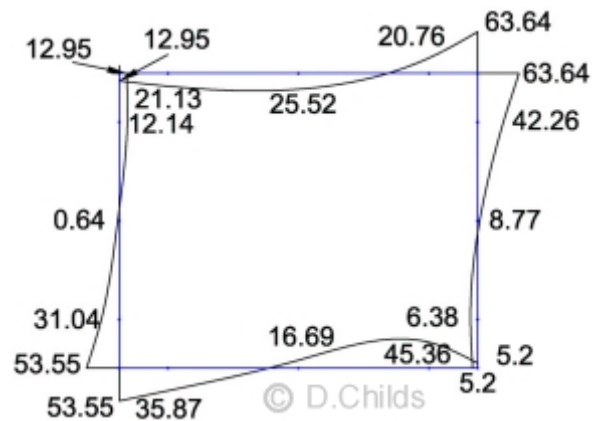
SLS Moments ($K_{\max} = 1.107$)

The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.



EQU Moments ($K_{\max} = 0.92$)

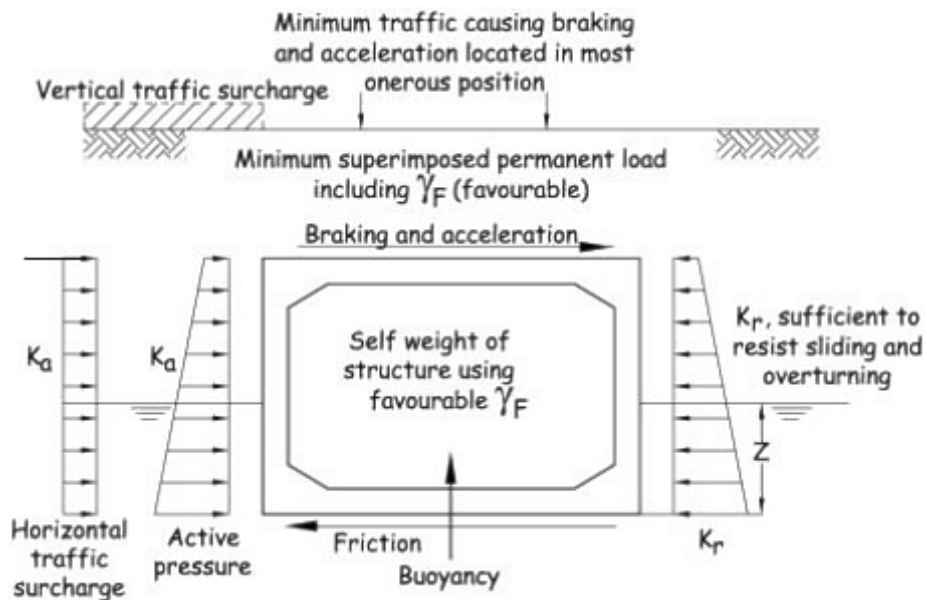
The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.



STR/GEO Comb. 1 Moments ($K_{\max} = 0.76$)

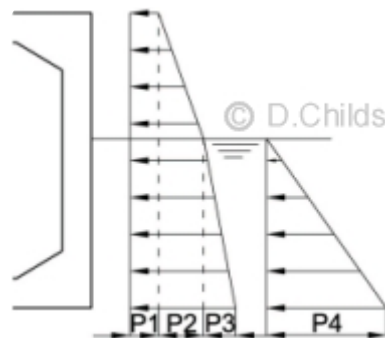
The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.

Load Case Table B.6



This is similar to Table B.5 loading with the addition of buoyancy effects due to ground water.

Pressures on walls:



P1 = Surcharge pressure from fill above roof level.

P2 = Backfill pressure from soil above water table.

P3 = Submerged backfill pressure from soil below water table.

P4 = water pressure.

Initial calculations will be based on using $K_{\max}$ values from Table B.5 for passive earth pressures. K_r values can then be calculated by proportion in order to achieve stability of the structure.

Sliding at SLS:

$\gamma_F = 1.0$ for all elements.

Use $K_r = 0.6$ for initial passive earth pressure coefficient.

Minimum surcharge from road construction = $0.6 \times 4.8 = 2.88 \text{ kN/m}$

Minimum surcharge from fill above roof level = 15.5kN/m

Active force from surcharge (P1) = $0.33 \times (2.88 + 15.5) \times 2.6 = 15.77\text{kN}$

Passive force from surcharge (P1) = $0.6 \times (2.88 + 15.5) \times 2.6 = 28.67\text{kN}$

Backfill pressure at water table level (P2) = $19 \times 1.3 = 24.7\text{kN/m}^2$

Active force from backfill from water table level up to roof = $0.33 \times 24.7 \times 1.3 / 2 = 5.30\text{kN}$

Active force from backfill surcharge from water table level up to roof = $0.33 \times 24.7 \times 1.3 = 10.60\text{kN}$

Submerged backfill pressure at foundation level (P3) = $(19 - 10) \times 1.3 = 11.7\text{kN/m}^2$

Active force from submerged backfill = $0.33 \times 11.7 \times 1.3 / 2 = 2.51\text{kN}$

Active force from ground water (P4) = $10 \times 1.3 \times 1.3 / 2 = 8.45\text{kN}$

Passive force from backfill from water table level up to roof = $0.6 \times 24.7 \times 1.3 / 2 = 9.63\text{kN}$

Passive force from backfill surcharge from water table level up to roof = $0.6 \times 24.7 \times 1.3 = 19.27\text{kN}$

Passive force from submerged backfill = $0.6 \times 11.7 \times 1.3 / 2 = 4.56\text{kN}$

Passive force from ground water (P4) = $10 \times 1.3 \times 1.3 / 2 = 8.45\text{kN}$

Active force from traffic surcharge line load = $0.33 \times 55 = 18.15\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.33 \times 30 \times 2.6 = 25.74\text{kN}$

Braking and acceleration force for LM3 traffic = 23.6kN

Total active force with LM3 traffic = $15.77 + 5.30 + 10.60 + 2.51 + 8.45 + 18.15 + 25.74 + 23.6 = 110.12\text{kN}$

Total passive force = $28.67 + 9.63 + 19.27 + 4.56 + 8.45 = 70.58\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $0.6 \times 4.8 \times 3.1 = 8.93\text{kN}$

Fill on roof = $15.5 \times 3.1 = 48.05\text{kN}$

Self weight of concrete = 76.5kN

LM3 traffic = 65.8kN

Buoyancy force = $1.3 \times 3.1 \times 10 = 40.3\text{kN}$

$V'_d = 8.93 + 48.05 + 76.5 + 65.8 - 40.3 = 158.98\text{kN}$

Maximum $R_d = 158.98 \times \tan 32^\circ = 99\text{kN}$

Friction required = $110.12 - 70.58 = 40 < 99 \therefore$ sliding can be resisted using $K_r = 0.6$
 $\therefore$ not critical.

Sliding at EQU:

Use $K_r = 0.6$ for initial passive earth pressure coefficient.

Active force from surcharge = $0.44 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 19.98\text{kN}$

Passive force from surcharge = $0.6 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 27.24\text{kN}$

Active force from backfill from water table level up to roof = $0.44 \times 0.95 \times 24.7 \times 1.3 / 2 = 6.71\text{kN}$

Active force from backfill surcharge from water table level up to roof = $0.44 \times 0.95 \times 24.7 \times 1.3 = 13.42\text{kN}$

Active force from submerged backfill = $0.44 \times 0.95 \times 11.7 \times 1.3 / 2 = 3.18\text{kN}$

Active force from ground water (P4) = $10 \times 0.95 \times 1.3 \times 1.3 / 2 = 8.03\text{kN}$

Passive force from backfill from water table level up to roof = $0.6 \times 0.95 \times 24.7 \times 1.3 / 2 = 9.15\text{kN}$

Passive force from backfill surcharge from water table level up to roof = $0.6 \times 0.95 \times 24.7 \times 1.3 = 18.30\text{kN}$

Passive force from submerged backfill = $0.6 \times 0.95 \times 11.7 \times 1.3 / 2 = 4.33\text{kN}$

Passive force from ground water (P4) = $10 \times 0.95 \times 1.3 \times 1.3 / 2 = 8.03\text{kN}$

Active force from traffic surcharge line load = $0.37 \times 1.35 \times 55 = 27.47\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.37 \times 1.35 \times 30 \times 2.6 = 38.96\text{kN}$

Braking and acceleration force for LM3 traffic = $1.35 \times 23.6 = 31.9\text{kN}$

Total active force with LM3 traffic = $19.98 + 6.71 + 13.42 + 3.18 + 8.03 + 27.47 + 38.96 + 31.9 = 149.65\text{kN}$

Total passive force = $27.24 + 9.15 + 18.3 + 4.33 + 8.03 = 67.05\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $0.95 \times 8.93 = 8.48\text{kN}$

Fill on roof = $0.95 \times 48.05 = 45.65\text{kN}$

Self weight of concrete = $0.95 \times 76.5 = 72.68\text{kN}$

LM3 traffic = $1.35 \times 65.8 = 88.83\text{kN}$

Buoyancy force = $1.3 \times 3.1 \times 1.05 \times 10 = 42.32\text{kN}$

$V'_d = 8.48 + 45.65 + 72.68 + 88.83 - 42.32 = 173.32\text{kN}$

Maximum $R_d = 173.32 \times \tan 29.6^\circ = 98.46\text{kN}$

Friction required = $149.65 - 67.05 = 83 < 98 \therefore$ sliding can be resisted using $K_r = 0.6$
 $\therefore$ not critical.

Sliding at ULS comb.1:

Use $K_r = 0.72$ for initial passive earth pressure coefficient.

Active force from surcharge = $0.40 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 18.16\text{kN}$

Passive force from surcharge = $0.72 \times 0.95 \times (2.88 + 15.5) \times 2.6 = 32.69\text{kN}$

Active force from backfill from water table level up to roof = $0.40 \times 0.95 \times 24.7 \times 1.3 / 2 = 6.10\text{kN}$

Active force from backfill surcharge from water table level up to roof = $0.40 \times 0.95 \times 24.7 \times 1.3 = 12.20\text{kN}$

Active force from submerged backfill = $0.40 \times 0.95 \times 11.7 \times 1.3 / 2 = 2.89\text{kN}$

Active force from ground water (P4) = $10 \times 0.95 \times 1.3 \times 1.3 / 2 = 8.03\text{kN}$

Passive force from backfill from water table level up to roof = $0.72 \times 0.95 \times 24.7 \times 1.3 / 2 = 10.98\text{kN}$

Passive force from backfill surcharge from water table level up to roof = $0.72 \times 0.95 \times 24.7 \times 1.3 = 21.96\text{kN}$

Passive force from submerged backfill = $0.72 \times 0.95 \times 11.7 \times 1.3 / 2 = 5.20\text{kN}$

Passive force from ground water (P4) = $10 \times 0.95 \times 1.3 \times 1.3 / 2 = 8.03\text{kN}$

Active force from traffic surcharge line load = $0.33 \times 1.35 \times 55 = 24.50\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.33 \times 1.35 \times 30 \times 2.6 = 34.75\text{kN}$

Braking and acceleration force for LM3 traffic = $1.35 \times 23.6 = 31.9\text{kN}$

Total active force with LM3 traffic = $18.16 + 6.1 + 12.2 + 2.89 + 8.03 + 24.5 + 34.75 + 31.9 = 138.53\text{kN}$

Total passive force = $32.69 + 10.98 + 21.96 + 5.2 + 8.03 = 78.86\text{kN}$

Active force > Passive force $\therefore$ friction under the base is required to stop sliding.

Vertical load on foundation:

Road construction = $0.95 \times 8.93 = 8.48\text{kN}$

Fill on roof = $0.95 \times 48.05 = 45.65\text{kN}$

Self weight of concrete = $0.95 \times 76.5 = 72.68\text{kN}$

LM3 traffic = $1.35 \times 65.8 = 88.83\text{kN}$

Buoyancy force = $1.3 \times 3.1 \times 1.20 \times 10 = 48.36\text{kN}$

$V'_d = 8.48 + 45.65 + 72.68 + 88.83 - 48.36 = 167.28\text{kN}$

Maximum $R_d = 167.28 \times \tan 32^\circ = 104.53\text{kN}$

Friction required = $138.53 - 78.86 = 60 < 105 \therefore$ sliding can be resisted using $K_r = 0.72 \therefore$ not critical.

Sliding at ULS comb.2

Use $K_r = 0.84$ for initial passive earth pressure coefficient.

Active force from surcharge = $0.49 \times 1.0 \times (2.88 + 15.5) \times 2.6 = 23.42\text{kN}$

Passive force from surcharge = $0.84 \times 1.0 \times (2.88 + 15.5) \times 2.6 = 40.14\text{kN}$

Active force from backfill from water table level up to roof = $0.49 \times 1.0 \times 24.7 \times 1.3 / 2 = 7.87\text{kN}$

Active force from backfill surcharge from water table level up to roof = $0.49 \times 1.0 \times 24.7 \times 1.3 = 15.73\text{kN}$

Active force from submerged backfill = $0.49 \times 1.0 \times 11.7 \times 1.3 / 2 = 3.73\text{kN}$

Active force from ground water (P4) = $10 \times 1.0 \times 1.3 \times 1.3 / 2 = 8.45\text{kN}$

Passive force from backfill from water table level up to roof = $0.84 \times 1.0 \times 24.7 \times 1.3 / 2 = 13.49\text{kN}$

Passive force from backfill surcharge from water table level up to roof = $0.84 \times 1.0 \times 24.7 \times 1.3 = 26.97\text{kN}$

Passive force from submerged backfill = $0.84 \times 1.0 \times 11.7 \times 1.3 / 2 = 6.39\text{kN}$

Passive force from ground water (P4) = $10 \times 1.0 \times 1.3 \times 1.3 / 2 = 8.45\text{kN}$

Active force from traffic surcharge line load = $0.41 \times 1.15 \times 55 = 25.93\text{kN}$

Active force from LM3 traffic UDL surcharge = $0.41 \times 1.15 \times 30 \times 2.6 = 36.78\text{kN}$
 Braking and acceleration force for LM3 traffic = $1.15 \times 23.6 = 27.1\text{kN}$
 Total active force with LM3 traffic = $23.42 + 7.87 + 15.73 + 3.73 + 8.45 + 25.93 + 36.78 + 27.1 = 149.01\text{kN}$
 Total passive force = $40.14 + 13.49 + 26.97 + 6.39 + 8.45 = 95.44\text{kN}$
 Active force > Passive force $\therefore$ friction under the base is required to stop sliding.
 Vertical load on foundation:
 Road construction = $1.0 \times 8.93 = 8.93\text{kN}$
 Fill on roof = $1.0 \times 48.05 = 48.05\text{kN}$
 Self weight of concrete = $1.0 \times 76.5 = 76.5\text{kN}$
 LM3 traffic = $1.15 \times 65.8 = 75.67\text{kN}$
 Buoyancy force = $1.3 \times 3.1 \times 1.0 \times 10 = 40.3\text{kN}$
 $V'_d = 8.93 + 48.05 + 76.5 + 75.67 - 40.3 = 168.85\text{kN}$
 Maximum $R_d = 168.85 \times \tan 26.6^\circ = 84.55\text{kN}$
 Friction required = $149.01 - 95.44 = 54 < 85 \therefore$ sliding can be resisted using $K_f = 0.84 \therefore$ not critical.

Overturning:

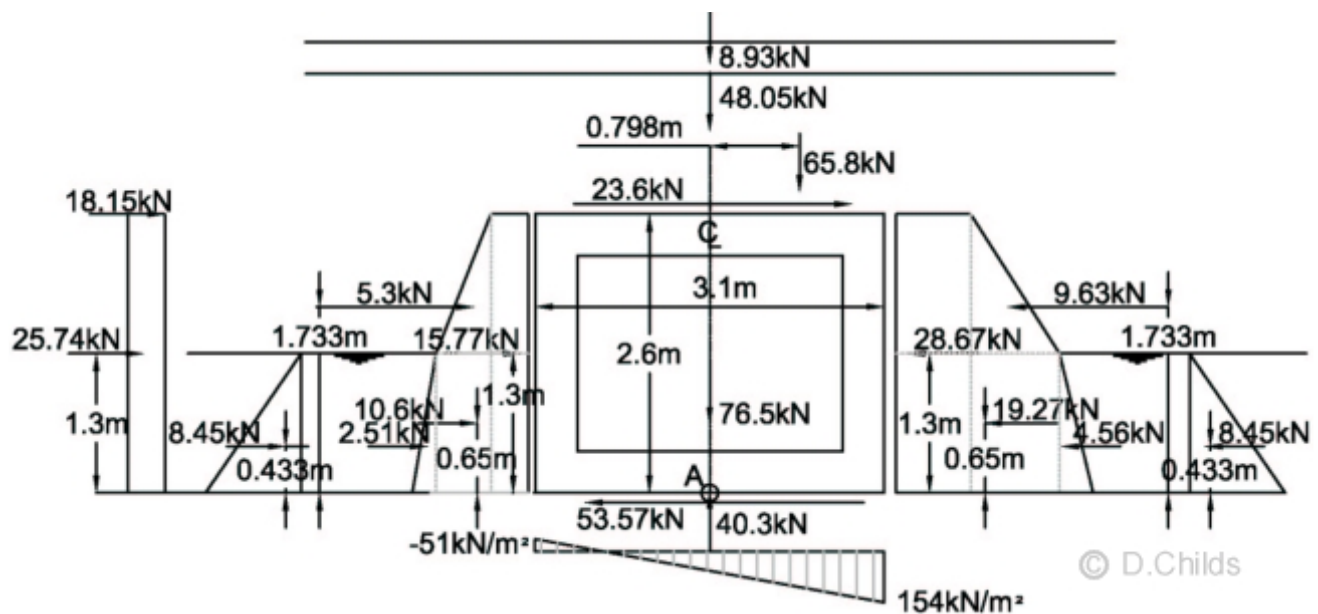
Stability against overturning is found by considering the eccentricity of loads about the centre of the foundation.

Consider SLS bearing pressure under base:

Total vertical load = $8.93 + 48.05 + 65.8 + 76.5 - 40.3 = 158.98\text{kN}$
 Taking moments about the centre of the base (point A in diagram below):
 (Ground water pressures have no net moment about A)
 Moment = $2.6 \times (18.15 + 23.6) + 1.733 \times (5.3 - 9.63) + 1.3 \times (25.74 + 15.77 - 28.67) + 0.65 \times (10.6 - 19.27) + 0.433 \times (2.51 - 4.56) + 65.8 \times 0.798 = 163.72\text{kNm}$
 Eccentricity $e = M / V = 163.72 / 158.98 = 1.03\text{m}$.
 No uplift will occur if $e < \text{base length} / 6 = 3.1 / 6 = 0.517 < 1.03\text{m} \therefore$ Fail.
 SLS pressures = $V/A \pm M / Z = 158.98 / (1.0 \times 3.1) \pm 163.72 / (1.0 \times 3.1^2 / 6) = 51.28 \pm 102.22 = 154 \text{ and } -51 \text{ kN/m}^2$.

Note 3 in Table B.6 says $K_{\max}$ may be increased to maintain stability, but check movement.

Active moment = $2.6 \times (18.15 + 23.6) + 1.733 \times 5.3 + 1.3 \times (25.74 + 15.77) + 0.65 \times 10.6 + 0.433 \times 2.51 + 65.8 \times 0.798 = 232.16\text{kNm}$
 Passive moment (with $K_f = 0.6$) = $1.733 \times 9.63 + 1.3 \times 28.67 + 0.65 \times 19.27 + 0.433 \times 4.56 = 68.46\text{kNm}$
 To avoid uplift $V/A - M/Z > 0$
 So $M < V \times Z / A = 158.98 \times 3.1^2 / (6 \times 3.1) = 82.14\text{kNm}$
 Passive moment required = $232.16 - 82.14 = 150.02\text{kNm}$



Bearing Pressure under Base at SLS with $K_r = 0.6$

Maximum bearing pressure = $158.98 / 3.1 + 82.14 \times 6 / 3.1^2 = 103\text{kN/m}^2$

K_r required = $0.6 \times 150.02 / 68.46 = 1.315$

Revised passive surcharge force = $28.67 \times 1.315 / 0.6 = 62.84\text{kN}$

Revised passive backfill force (P_2) = $9.63 \times 1.315 / 0.6 = 21.11\text{kN}$

Revised passive backfill surcharge force (P_2) = $19.27 \times 1.315 / 0.6 = 42.23\text{kN}$

Revised submerged passive backfill force (P_3) = $4.56 \times 1.315 / 0.6 = 9.99\text{kN}$

Check for movement to develop $K_r = 1.32$:

From BS EN 1997-1:2004 Figure C.2.1 (with ϕ' for 6N backfill = 35°) $K_p = 3.7$

Proportion of K_p required = $1.32 / 3.7 = 0.357 (< 0.5)$

PD 6694-1:2011 Clause 7.5 says movement (v/h) varies linearly from zero to v_2/h as K increases from K_0 to $0.5K_p$

$0.5K_p = 3.7 / 2 = 1.85$

Tables B.1 and B.2 quote K_0 as 0.5 at SLS

Proportion of v_2/h movement = $(1.32 - 0.5) / (1.85 - 0.5) = 0.607$

From BS EN 1997-1:2004 Table C.2 for dense soil the range of v_2/h for rotational movement = 1.1% to 2.0%

Maximum movement = $2\% \times 0.607 \times 2.6\text{m} = 32\text{mm}$

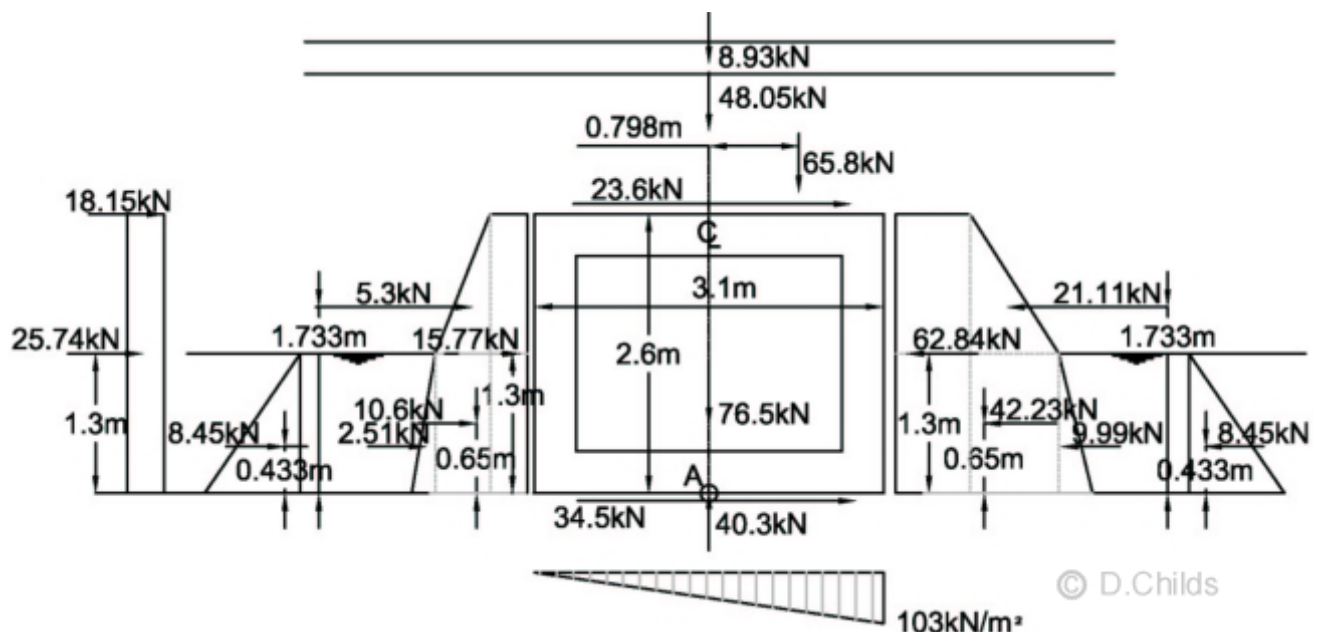
This movement would need to be checked for impact on adjacent services etc. and be accepted by the approving authority.

For lateral stability under increased passive force:

Active horizontal force = $15.77 + 5.3 + 10.6 + 2.51 + 18.15 + 25.74 + 23.6 + 8.45 = 110.12\text{kN}$

Passive horizontal force = $62.84 + 21.11 + 42.23 + 9.99 + 8.45 = 144.62\text{kN}$

Although the passive force is greater than the active force it would still be prudent to check the structure for the increased loading.



Bearing Pressure under Base at SLS with $K_r = 1.32$.

Check for eccentricity of loads at ULS:

Consider EQU bearing pressure under base:

Total vertical load = $8.48 + 45.65 + 65.8 \times 1.35 + 72.68 - 42.32 = 173.32\text{kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (27.47 + 31.9) + 1.733 \times (6.71 - 9.15) + 1.3 \times (38.96 + 19.98 - 27.24) + 0.65 \times (13.42 - 18.3) + 0.433 \times (3.18 - 4.33) + 88.83 \times 0.798 = 258.56\text{kNm}$

Eccentricity $e = M / V = 258.56 / 173.32 = 1.492\text{m}$.

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.03 < 1.492\text{m} \therefore$ Fail.

Moment required to limit e to $1.033\text{m} = 1.033 \times 173.32 = 179.04\text{kNm}$

Moment from passive forces (using $K_r = 0.6$) = $1.733 \times 9.15 + 1.3 \times 27.24 + 0.65 \times 18.3 + 0.433 \times 4.33 = 65.04$

K_r required to limit $e = 0.6 \times (65.04 + 258.56 - 179.04) / 65.04 = 1.33 (> 1.32 \text{ for SLS} \therefore \text{critical})$

The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 173.32 / (3.1 - 2 \times 1.033) = 167.6\text{kN/m}^2$.

Revised passive force from surcharge = $27.24 \times 1.33 / 0.6 = 60.38\text{kN}$

Revised passive backfill force (P_2) = $9.15 \times 1.33 / 0.6 = 20.28\text{kN}$

Revised passive backfill surcharge force (P_2) = $18.3 \times 1.33 / 0.6 = 40.57\text{kN}$

Revised submerged passive backfill force (P_3) = $4.33 \times 1.33 / 0.6 = 9.60\text{kN}$

Total active force with LM3 traffic = 149.65kN

Total passive force = $60.38 + 20.28 + 40.57 + 9.6 + 8.03 = 138.86\text{kN}$

Friction force on base to maintain stability = $149.65 - 138.86 = 10.79\text{kN} < 98\text{kN} \therefore \text{O.K.}$

Check for movement to develop $K_r = 1.33$:

From BS EN 1997-1:2004 Figure C.2.1 (with ϕ' for 6N backfill = 35°) $K_p = 3.7$

Proportion of K_p required = $1.33 / 3.7 = 0.359 (< 0.5)$

PD 6694-1:2011 Clause 7.5 says movement (v/h) varies linearly from zero to v_2/h as K increases from K_0 to $0.5K_p$

$$0.5K_p = 3.7 / 2 = 1.85$$

Tables B.1 and B.2 quote K_0 as 0.5 at SLS

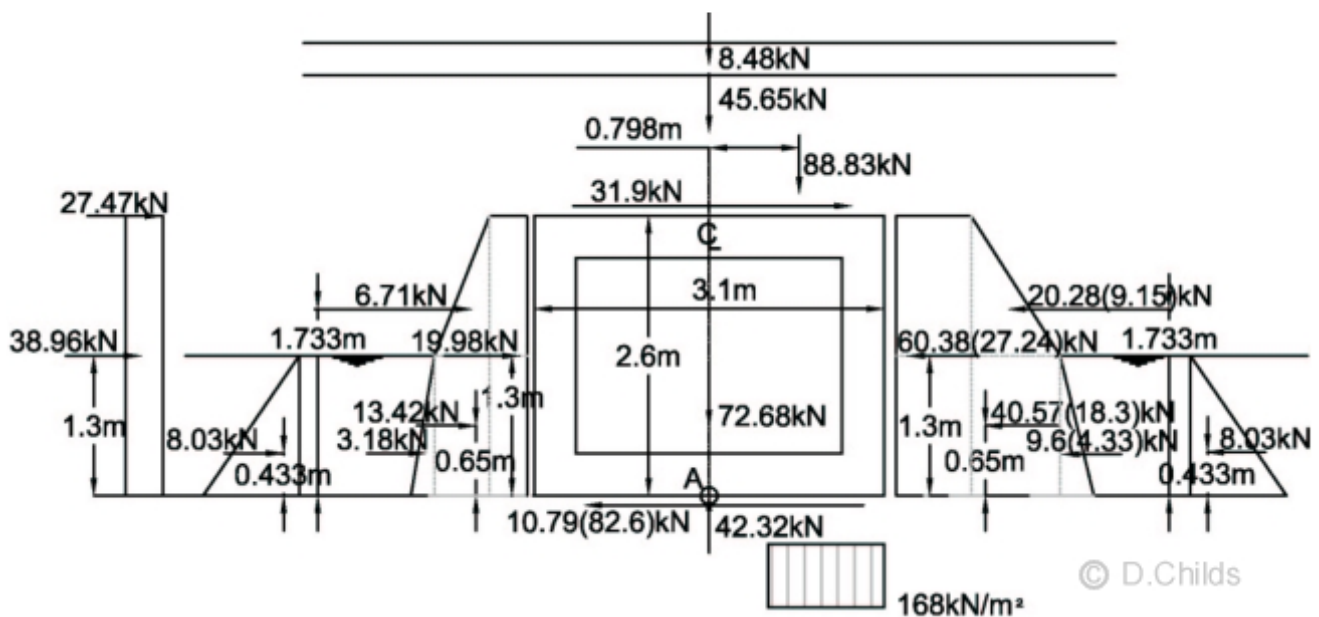
$$\text{Proportion of } v_2/h \text{ movement} = (1.33 - 0.5) / (1.85 - 0.5) = 0.615$$

From BS EN 1997-1:2004 Table C.2 for dense soil the range of v_2/h for rotational movement = 1.1% to 2.0%

$$\text{Maximum movement} = 2\% \times 0.615 \times 2.6\text{m} = 32\text{mm}$$

This movement would need to be checked for impact on adjacent services etc. and be accepted by the approving authority.

The structure will need to be assessed for the increased loading.



Bearing Pressure under Base at EQU using $K_r = 1.33$ (Values in brackets using $K_r = 0.6$)

Consider STR/GEO (Comb 1) bearing pressure under base:

$$\text{Total vertical load} = 8.48 + 45.65 + 65.8 \times 1.35 + 72.68 - 48.36 = 167.28\text{kN}$$

Taking moments about the centre of the base (point A in diagram below):

$$\begin{aligned} \text{Moment} = & 2.6 \times (24.5 + 31.9) + 1.733 \times (6.1 - 10.98) + 1.3 \times (34.75 + 18.16 - 32.69) \\ & + 0.65 \times (12.2 - 21.96) + 0.433 \times (2.89 - 5.2) + 88.83 \times 0.798 = 228.01\text{kNm} \end{aligned}$$

$$\text{Eccentricity } e = M / V = 228.01 / 167.28 = 1.363\text{m.}$$

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.033 < 1.363\text{m} \therefore$ Fail.

$$\text{Moment required to limit } e \text{ to } 1.033\text{m} = 1.033 \times 167.28 = 172.8\text{kNm}$$

$$\begin{aligned} \text{Moment from passive forces (using } K_r = 0.72) = & 1.733 \times 10.98 + 1.3 \times 32.69 + 0.65 \times \\ & 21.96 + 0.433 \times 5.2 = 78.05 \end{aligned}$$

$$\begin{aligned} K_r \text{ required to limit } e = & 0.72 \times (78.05 + 228.01 - 172.8) / 78.05 = 1.229 (< 1.33 \text{ for EQU} \\ \therefore & \text{not critical}) \end{aligned}$$

The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 167.28 / (3.1 - 2 \times 1.033) = 161.8 \text{ kN/m}^2$.

Revised passive force from surcharge = $32.69 \times 1.229 / 0.72 = 55.8 \text{ kN}$

Revised passive backfill force (P2) = $10.98 \times 1.229 / 0.72 = 18.74 \text{ kN}$

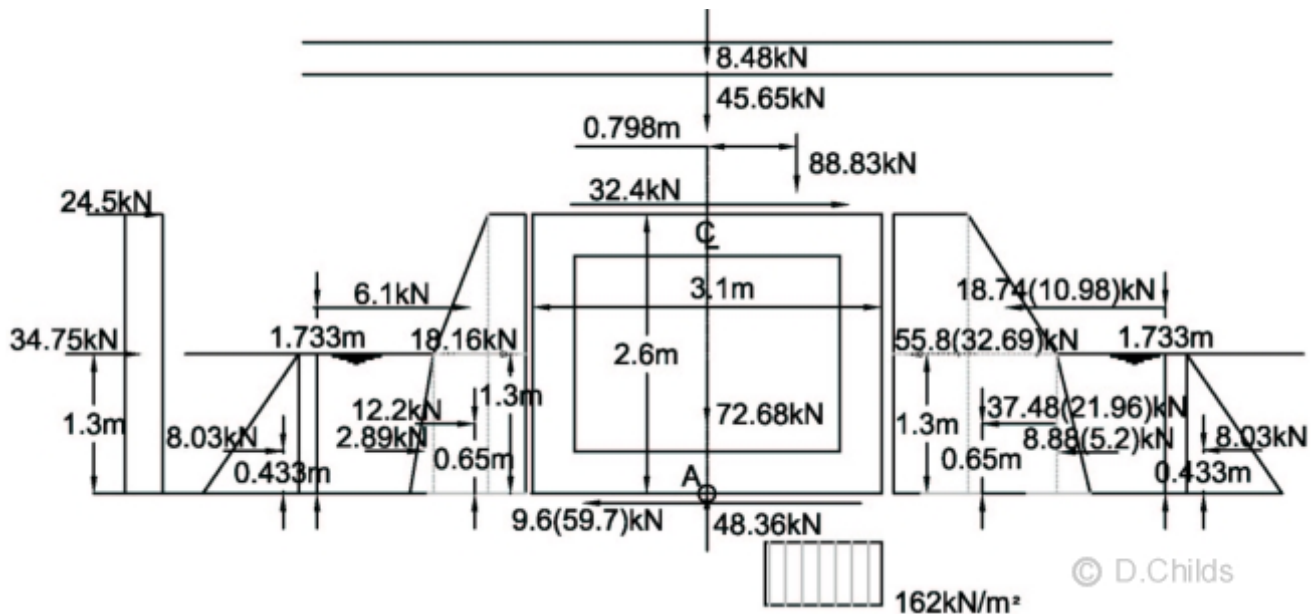
Revised passive backfill surcharge force (P2) = $21.96 \times 1.229 / 0.72 = 37.48 \text{ kN}$

Revised submerged passive backfill force (P3) = $5.2 \times 1.229 / 0.72 = 8.88 \text{ kN}$

Total active force with LM3 traffic = 138.53 kN

Total passive force = $55.8 + 18.74 + 37.48 + 8.88 + 8.03 = 128.9 \text{ kN}$

Friction force on base to maintain stability = $138.53 - 128.9 = 9.6 \text{ kN} < 105 \text{ kN} \therefore \text{O.K.}$



Bearing Pressure under Base at STR/GEO (Comb 1) using $K_r = 1.23$ (Values in brackets using $K_r = 0.72$)

Consider STR/GEO (Comb 2) bearing pressure under base:

Total vertical load = $8.93 + 48.05 + 65.8 \times 1.15 + 76.5 - 40.3 = 168.85 \text{ kN}$

Taking moments about the centre of the base (point A in diagram below):

Moment = $2.6 \times (25.93 + 27.1) + 1.733 \times (7.87 - 13.49) + 1.3 \times (36.78 + 23.42 - 40.14) + 0.65 \times (15.73 - 26.97) + 0.433 \times (3.73 - 6.39) + 75.67 \times 0.798 = 206.14 \text{ kNm}$

Eccentricity $e = M / V = 206.14 / 168.85 = 1.22 \text{ m}$.

Check whether eccentricity exceeds base length / 3 = $3.1 / 3 = 1.033 < 1.22 \text{ m} \therefore \text{Fail}$.

Moment required to limit e to $1.033 \text{ m} = 1.033 \times 168.85 = 174.42 \text{ kNm}$

Moment from passive forces (using $K_r = 0.84$) = $1.733 \times 13.49 + 1.3 \times 40.14 + 0.65 \times 26.97 + 0.433 \times 6.39 = 95.86$

K_r required to limit $e = 0.84 \times (95.86 + 206.14 - 174.42) / 95.86 = 1.118 (< 1.33 \text{ for EQU} \therefore \text{not critical})$

The bearing pressure diagram will be rectangular at ULS so pressure = $V / (L - 2e) = 168.85 / (3.1 - 2 \times 1.033) = 163.3 \text{ kN/m}^2$.

Revised passive force from surcharge = $40.14 \times 1.118 / 0.84 = 53.42 \text{ kN}$

Revised passive backfill force (P2) = $13.49 \times 1.118 / 0.84 = 17.95\text{kN}$

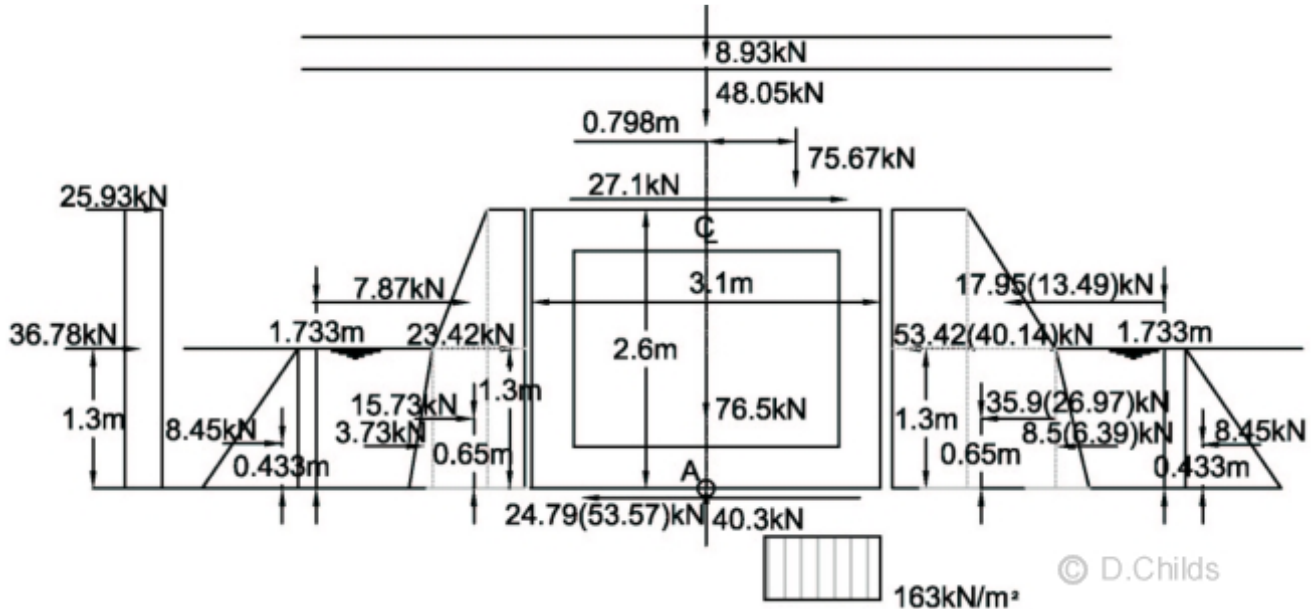
Revised passive backfill surcharge force (P2) = $26.97 \times 1.118 / 0.84 = 35.9\text{kN}$

Revised submerged passive backfill force (P3) = $6.39 \times 1.118 / 0.84 = 8.5\text{kN}$

Total active force with LM3 traffic = 149.01kN

Total passive force = $53.42 + 17.95 + 35.9 + 8.5 + 8.45 = 124.22\text{kN}$

Friction force on base to maintain stability = $149.01 - 124.22 = 24.79\text{kN} < 85\text{kN} \therefore \text{O.K.}$



Bearing Pressure under Base at STR/GEO (Comb 2) using $K_r = 1.12$ (Values in brackets using $K_r = 0.84$)

Check Bearing Resistance from BS EN 1997-1 Clause D.4

Equation (D.2): $R/A' = c'N_c b_c s_c i_c + q'N_q b_q s_q i_q + 0.5\gamma_d B'N_\gamma b_\gamma s_\gamma i_\gamma$

$c' = 0$ (cohesionless soil)

$q' = 1.5 \times 19 = 28.5 \text{ kN/m}^2$ (overburden pressure)

$\gamma' = 19 \text{ kN/m}^3$ (weight density of foundation material)

$N_q = e^{\pi \tan \phi'_d \tan^2(45 + \phi'_d/2)}$

$N_\gamma = 2(N_q - 1) \tan \phi'$, where $\delta \geq \phi' / 2$ (rough base)

$b_q = b_\gamma = (1 - \alpha \tan \phi'_d)$

Inclination $\alpha = 0$

$s_q = 1 + (B' / L') \sin \phi'_d$

$s_\gamma = 1 - 0.3(B' / L')$

$m = m_B = [2 + (B' / L')] / [1 + (B' / L')]$ when H acts in the direction of B'

$i_q = [1 - H / (V + A'c'_d \cot \phi'_d)]^m$

$i_\gamma = [1 - H / (V + A'c'_d \cot \phi'_d)]^{m+1}$

From Figure D.1:

$L' = 20.6\text{m}$ (transverse width of foundation)

$B = 3.1\text{m}$ $B' = B - 2e_B$

For 1m strip design $A' = 1.0 \times B' = B'$

	SLS	EQU	STR/GEO Comb1	STR/GEO Comb2
e_B	0.517	1.033	1.033	1.033
$A' \text{ \& } B'$	$3.1 - 2 \times 0.517 = 2.067$	$3.1 - 2 \times 1.033 = 1.033$	$3.1 - 2 \times 1.033 = 1.033$	$3.1 - 2 \times 1.033 = 1.033$
$\gamma_d = \gamma' \times \gamma_{G;\text{sup}}$	$19.0 \times 1.0 = 19.0$	$19.0 \times 0.95 = 18.05$	$19.0 \times 0.95 = 18.05$	$19.0 \times 1.0 = 19.0$
ϕ'_d	$\tan^{-1}(\tan 32 / 1.0) = 32^\circ$	$\tan^{-1}(\tan 32 / 1.1) = 29.6^\circ$	$\tan^{-1}(\tan 32 / 1.0) = 32^\circ$	$\tan^{-1}(\tan 32 / 1.25) = 26.6^\circ$
N_q	23.18	17.59	23.18	12.64
N_γ	27.72	18.85	27.72	11.66
$b_q \text{ \& } b_\gamma$	1.0	1.0	1.0	1.0
s_q	1.053	1.025	1.027	1.023
s_γ	0.97	0.985	0.985	0.985
m	1.909	1.952	1.952	1.952
Net H_d	$144.62 - 110.12 = 34.5$	$149.65 - 138.86 = 10.79$	$138.53 - 128.9 = 9.63$	$149.01 - 124.22 = 24.79$
V_d	158.98	173.32	167.28	168.85
i_q	0.627	0.889	0.895	0.733
i_γ	0.491	0.837	0.845	0.626
$q'N_q b_q s_q i_q$	436	457	607	269

$0.5\gamma_d B' N_\gamma b_\gamma s_\gamma i_\gamma$	259	144	215	70
R / A'	695	601	822	339
V_d / A'	$158.98 / 2.066 = 77$	$173.32 / 1.033 = 168$	$167.28 / 1.033 = 162$	$168.85 / 1.033 = 163$
R / V_d	9.04	3.59	5.08	2.08

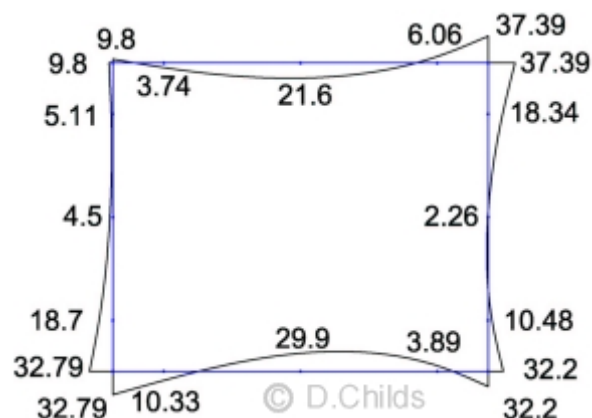
PD 6694-1:2011 Clause 5.2.2 says the serviceability limit state for settlement can be verified by ensuring that a "sufficiently low fraction of the ground strength is mobilized". This requirement can be deemed to be satisfied if the maximum pressure under a foundation at SLS does not exceed one third of the design resistance R/A' .

$$1/3 (R / A') = 695 / 3 = 233 \text{ kN/m}^2$$

Maximum bearing pressure at SLS = $103 \text{ kN/m}^2 < 233 \therefore$ settlement requirement can be deemed to be satisfied.

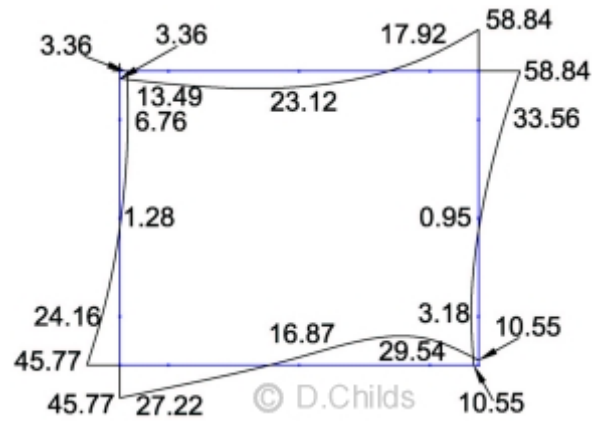
Minimum Bearing Resistance Factor = $2.08 > 1.0 \therefore$ O.K.

K_r had to be increased from the tabulated $K_{\max}$ values for all Limit States to maintain stability of the structure so we need to ensure that the structure is able to sustain the increased pressures. The plane frame analysis program is used to determine the additional load effects. Minor adjustments to the bearing pressures need to be made to minimise the magnitude of the artificial reactions at nodes 1 and 13.



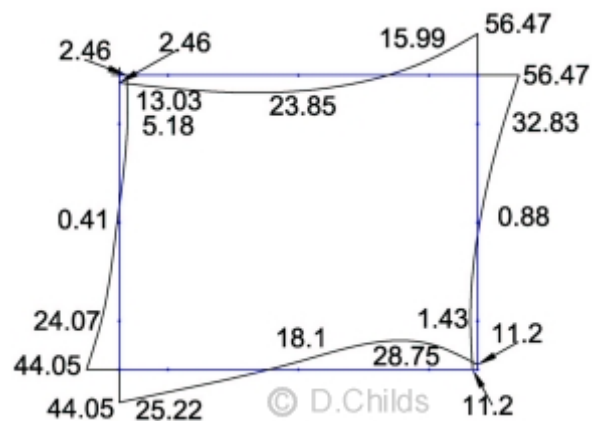
SLS Moments ($K_r = 1.32$)

The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.



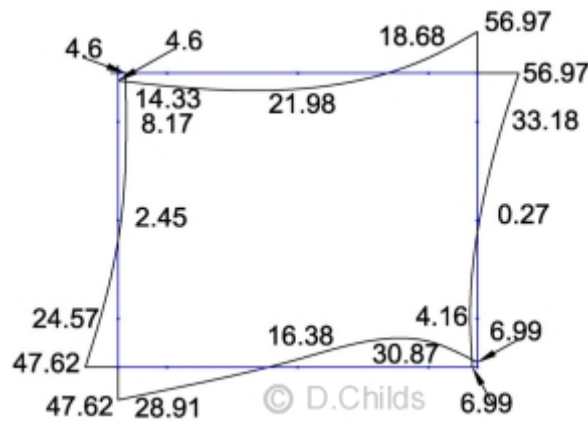
EQU Moments ($K_r = 1.33$)

The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.



STR/GEO Comb. 1 Moments ($K_r = 1.23$)

The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.

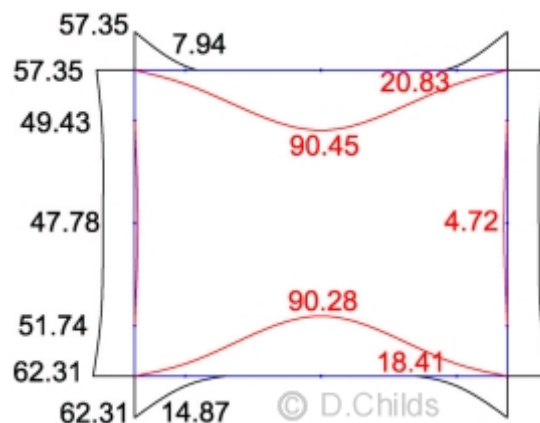


STR/GEO Comb. 2 Moments ($K_r = 1.12$)

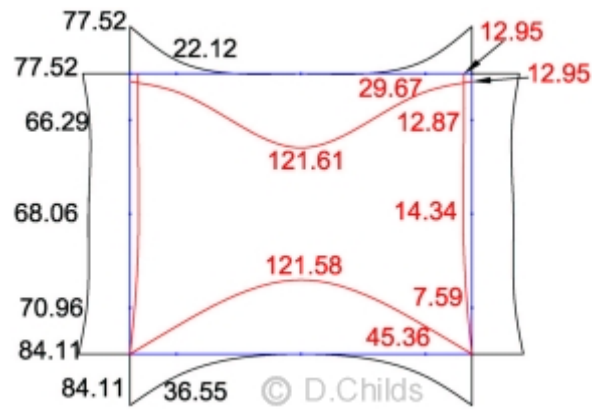
The loading was entered in [the plane frame spreadsheet](#) as a single load case. No allowance was made for the difference in short-term and long-term loading as this will only affect the values of the deflections and not the bending moment, shear and axial effects.

Reinforcement Design

Taking the maximum moments from the results of loading tables B.1 to B.6, and assuming all loading can be mirrored, an envelope is produced. Values shown in black are for moments which produce tension on the outside face whilst values in red are for tension of the inside face of the culvert. Only half the values are shown for clarity; the values are mirrored about the vertical axis.



Envelope of SLS Moments



Envelope of ULS Moments

Use C32/40 concrete to BS 8500.

Use Grade B500B reinforcement to BS 4449.

BS 8500-1

cl. A.2.1 Table A.1: Assume exposure Class XD2.

cl. A.3: Fixing tolerance for reinforcement $\Delta_c = 15\text{mm}$ for insitu concrete.

Table A.5: Nominal cover for C32/40 concrete = $45 + \Delta_c = 60\text{mm}$ with maximum water-cement ratio = 0.50 and minimum cement content of 340 kg/m^3 .

EN 1992-2

The reinforcement requirements can be determined by following the procedure in the ['Reinforced Concrete Deck'](#) example or by using a [simple spreadsheet](#).

Maximum shear and axial loads are obtained in a similar process to the moments.

Critical shear positions are located at d away from the face of the support. Nodes 2, 4, 6, 8, 10, 12, 14 and 16 were located in the plane frame model to obtain these values.

Maximum shear values obtained were:

Nodes 2 & 12 = 55kN

Nodes 4 & 10 = 61kN

Nodes 6 & 8 = 191kN

Nodes 14 & 16 = 202kN

All axial values were compressive with the exception of one load case with maximum water level inside the culvert which produced a tension of 1.4kN in the floor slab. Maximum axial values obtained were:

Walls = 262kN at top to 281kN at bottom

Roof = 102kN

Floor = 84kN

Maximum moments are at mid-span in the roof and floor; these were produced in Table B.3 loading. SLS moments need to be specified in terms of effects from quasi-permanent and short-term actions.

The SLS mid-span roof moment of 90.45kNm consists of 20.39kNm quasi-permanent loading and 70.06kNm short-term loading (load case D7).

The SLS mid-span floor moment of 90.28kNm consists of 27.35kNm quasi-permanent loading and 62.93kNm short-term loading (load case D7).

M_{qp} / M_{st} is 0.291 for the roof moment and 0.435 for the floor moment.

Using spreadsheet 304 suitable reinforcement for a 300mm thick slab is **16mm dia. reinforcement at 125mm centres** ($M_{ult} = 148\text{kNm} > 121.6\text{kNm}$, M_{sls} for characteristic combinations = $118\text{kNm} > 90.45\text{kNm}$, M_{sls} for quasi-permanent combinations = $82\text{kNm} > 27.35\text{kNm}$).

All sections in the structure can be designed for bending in a similar way.

Shear

Shear is considered at a distance d away from the support. Nodes on the plane frame model were positioned at these points.

Worst shear in floor slab at nodes 14 and 16 is 202kN

Spreadsheet 304 shows 16mm dia. reinforcement at 125mm centres ($A_s = 1608\text{mm}^2/\text{m}$) has a shear resistance of 302kN at distance d away from the support.

As the critical sections (d from support) are close to points of contraflexure then tension can occur both on the inside and outside faces of the structure. The longitudinal tensile steel to resist shear should therefore be provided on both faces.

Shear resistance in the walls can be checked using EN 1992-1-1 Clause 6.2.2 equation 6.2a:

$$V_{Rd,c} = [C_{Rd,c} k (100 \rho_1 f_{ck})^{1/3} + k_1 \sigma_{cp}] b_w d$$

Minimum axial compressive load N_{Ed} in wall at critical shear nodes 4 & 10 (load case table B.2 ULS Comb.1) = 39.75kN

$$A_c = 1000 \times 300 = 300 \times 10^3 \text{mm}^2$$

$$\sigma_{cp} = N_{Ed} / A_c < 0.2 f_{cd}$$

$$\sigma_{cp} = 39.75 / 300 = 0.133 \text{ N/mm}^2$$

$$f_{cd} = 18.133 \text{ N/mm}^2 \text{ (from spreadsheet 304)}$$

$$0.2 f_{cd} = 0.2 \times 18.133 = 3.627 > 0.133 \therefore \text{OK}$$

Try using B12@125 then

$$d = 300 - 60 - 12 / 2 = 234$$

$$k_1 = 0.15 \text{ from UK NA to BS EN 1992-1-1 Table NA.1}$$

$$k_1 \sigma_{cp} b_w d = 0.15 \times 0.133 \times 1000 \times 234 / 1000 = 4.7 \text{kN}$$

Maximum ULS moment in wall at nodes 2 and 12 = 70.96kNm

Using spreadsheet 304 to check the capacity of B12@125c/c (ULS moment capacity =

88kNm > 70.96) then the shear capacity at d from support taking account of the axial effects = $2 \times (124.99 + 4.7) = 259.38\text{kN} > 61\text{kN} \therefore \text{OK}$.

It should be noted that the minimum axial (Table B.2 comb.1) and shear (Table B.6 EQU) effects are not co-existing, however the calculation does show the reinforcement will be adequate. Not using co-existing values can lead to over designing the section and hence an uneconomical solution.

All sections in the structure can be designed for shear and axial effects in a similar way.

As there are no joints in the structure then it is important to check for early thermal crack control considering:

- i. The floor slab being cast onto blinding
- ii. The walls being cast onto the floor slab
- iii. The roof slab being cast onto the walls

This can be achieved by using [spreadsheet 306](#).

[Back to Buried Box Tutorial](#) | [Back to Tutorial Index](#)

Contact [David Childs](#)